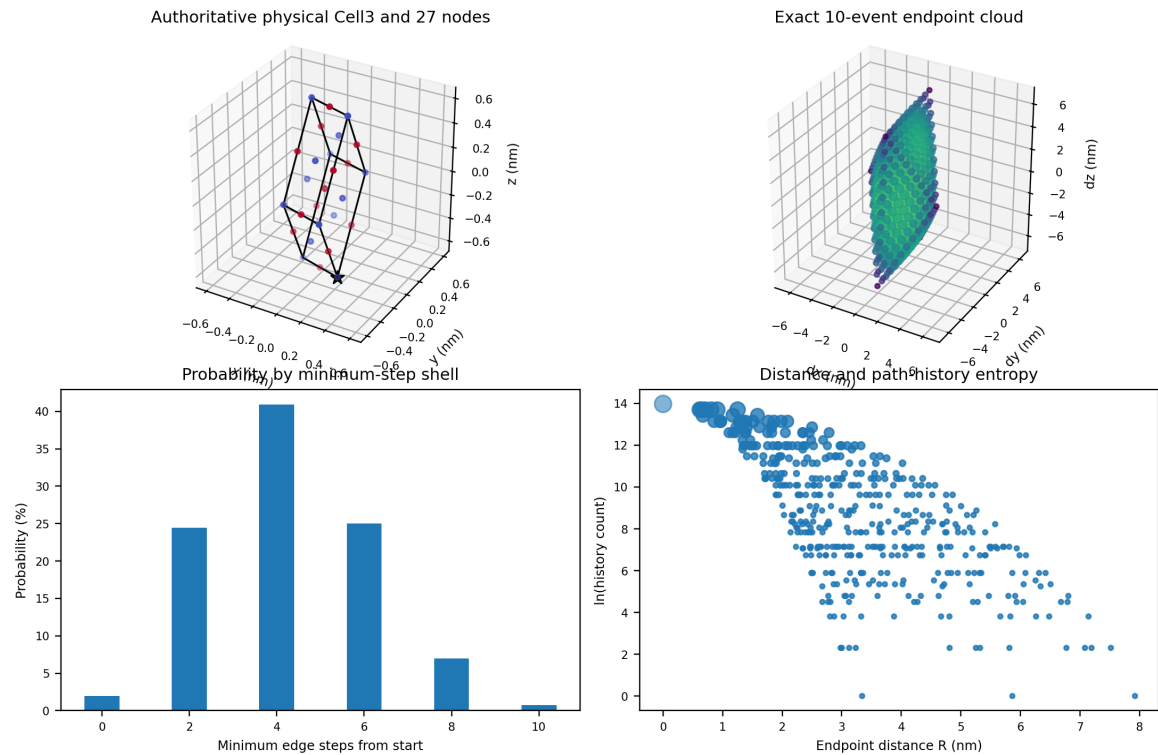


# K-FIELD 10-EVENT CELL3 PROPAGATION EXPERIMENT

Exact path enumeration, spatial probability, entropy fibers, and metastable-device interpretation

K-Field 10-event Cell3 propagation: geometry, probability, and entropy



**Figure 1. Title composite.** Four exact views generated from the authoritative physical Cell3 basis and the complete 10-event enumeration: the 27-node Cell3, the endpoint cloud, the minimum-step shell distribution, and path-history entropy versus physical distance.

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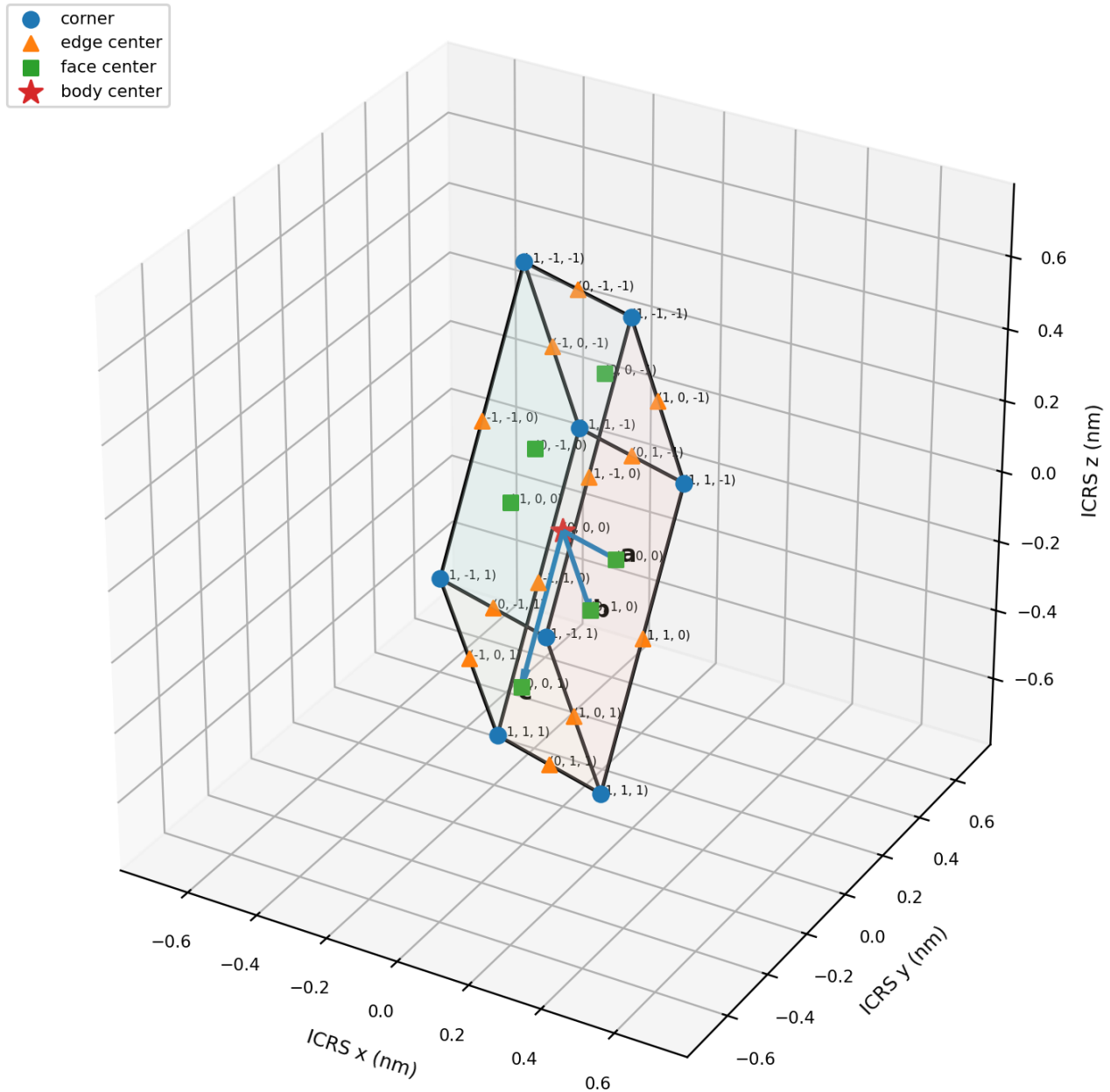
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# Frontispiece: authoritative physical Cell3, face planes, and all nodes

Dimensionally correct physical Cell3: six boundary planes, 12 edges, and all 27 nodes



**Figure 2. Authoritative physical Cell3.** The centered primitive cell is rendered directly from the locked nanometre basis. All six boundary faces, 12 cell edges, 27 address nodes, exact integer node labels, and the three ordered basis directions are shown in physical ICRS Cartesian dimensions.

The frontispiece establishes the geometry used throughout the experiment. The 27 special nodes are the half-step address set  $\{-1, 0, 1\}^3$ . The eight addresses with three nonzero components are corners; the 12 addresses with two nonzero components are edge centers; the six addresses with one nonzero component are face centers; and  $(0, 0, 0)$  is the body center. Their physical positions are one-half of the locked Cell3 basis acting on the integer address.

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## Abstract

This paper gives a self-contained exact propagation experiment in the volume-filling K-Field Cell3 address space. The experiment begins at the physical corner node with local address (1,1,1). Ten metastable transition events are then applied. Each event is assigned one of the six directed primitive edge moves +a, -a, +b, -b, +c, or -c. The baseline transition law is unbiased, so every directed move has probability 1/6. No Monte Carlo sampling is used: all  $6^{10} = 60,466,176$  ordered histories are enumerated exactly by dynamic programming and collected by their final integer displacement address.

The complete history ensemble terminates on 891 distinct corner nodes. The starting corner receives 1,172,556 histories and therefore has return probability 1.939193 percent. The mean endpoint displacement is zero, but the endpoint cloud is not spherical. The unequal and nonorthogonal Cell3 edges produce an elongated physical covariance ellipsoid with principal standard deviations 1.651006, 0.725471, and 0.599126 nm. The mean endpoint distance is 1.685874 nm, the root-mean-square distance is 1.900285 nm, and the largest reachable displacement is 7.923198 nm.

The exact history entropy is  $10 \ln 6 = 17.917595$  nats. Grouping histories only by destination leaves endpoint entropy 5.364728 nats and mean conditional path-history, or fiber, entropy 12.552867 nats. Thus approximately 70.06 percent of the ten-event information remains in the unresolved history fiber after the endpoint is known. This decomposition supplies a precise probabilistic interpretation of metastable K-Field observations: Cell3 geometry determines the transition paths, path multiplicity determines baseline occupancy, and directional transition bias deforms the predicted spatial distribution.

**Central result.** At a fixed event count and transition law, every destination address has an exactly determined probability and an exactly determined path-history entropy. Physical distance constrains the geometry of the endpoint, but distance alone does not uniquely determine probability or entropy. The complete directed displacement address does.

## Reader discipline and scope

The paper keeps four quantities distinct. Event count is the number of transitions executed. Minimum edge-step count is the shortest number of primitive edge transitions needed to connect the start and endpoint. Traversed physical path length is the sum of the actual physical edge lengths used by one history. Endpoint distance is the direct metric displacement  $\sqrt{d^T G d}$ . These quantities are related but are not interchangeable.

The exact unbiased distribution is the geometric null model. The later directional-bias example is a computed sensitivity demonstration using an explicitly declared probability vector; it is not presented as measured device data. In an actual metastable device, the six transition probabilities must be estimated from observed event counts under controlled device orientation, drive phase, temperature, and other experimental conditions.

## 1. K-Field Cell3 geometry and address space

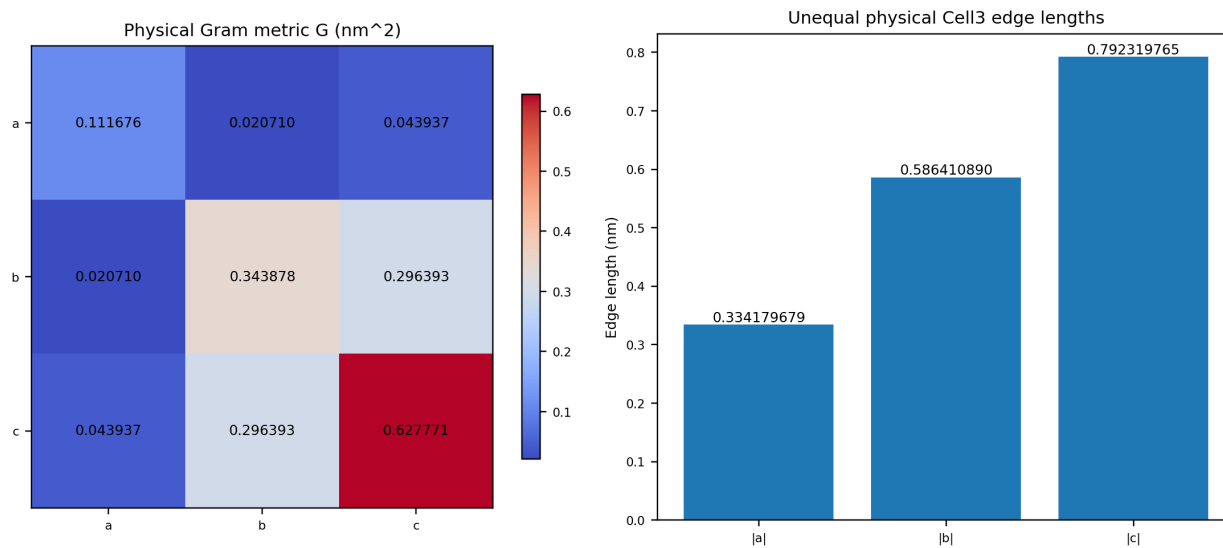
### 1.1 Ordered physical basis

The volume-filling geometry is generated by the ordered physical basis B. Its columns are the Cell3 edge vectors a, b, and c. A cell translated by integer address  $m=(m_a, m_b, m_c)$  has physical translation  $Bm$ . The same ordered basis therefore generates both the primitive cell and the complete corner-node lattice used in this experiment.

$$B = \begin{bmatrix} 0.3275012420558906, & 0.0353307758438284, & -0.01012931762672882, \\ -0.0548970078840064, & 0.2073198504401918, & -0.38978770236625987, \\ -0.0374875028947299, & -0.5474010853885081, & -0.68973440836726760 \end{bmatrix} \text{ nm}$$

| Edge | x component (nm)  | y component (nm)   | z component (nm)   | Length (nm)       |
|------|-------------------|--------------------|--------------------|-------------------|
| a    | 0.327501242055891 | -0.054897007884006 | -0.037487502894730 | 0.334179679059111 |
| b    | 0.035330775843828 | 0.207319850440192  | -0.547401085388508 | 0.586410890411141 |

| Edge | x component (nm)   | y component (nm)   | z component (nm)   | Length (nm)       |
|------|--------------------|--------------------|--------------------|-------------------|
| c    | -0.010129317626729 | -0.389787702366260 | -0.689734408367268 | 0.792319765042685 |



**Figure 3. Metric and edge-length ledger.** The physical Gram metric contains the six independent quadratic geometry channels. The edge-length panel shows why a fixed number of events does not imply a single traversed physical length.

## 1.2 Gram metric and endpoint distance

The direct Gram metric is  $G=B^T B$ . For any integer displacement address  $d$ , the direct physical endpoint displacement and its squared distance are

$$\begin{aligned}\Delta r &= B d, \\ Q(d) &= d^T G d, \\ R(d) &= \sqrt{Q(d)}.\end{aligned}$$

| row | a                 | b                 | c                 |
|-----|-------------------|-------------------|-------------------|
| a   | 0.111676057896050 | 0.020710333280606 | 0.043937235096220 |
| b   | 0.020710333280606 | 0.343877732392787 | 0.296392758961523 |
| c   | 0.043937235096220 | 0.296392758961523 | 0.627770610077296 |

The physical cell volume is  $0.117914891911745 \text{ nm}^3$ . The edge angles are  $\theta_{ab}=83.933483012 \text{ deg}$ ,  $\theta_{ac}=80.448130018 \text{ deg}$ , and  $\theta_{bc}=50.363231814 \text{ deg}$ . Because none of the edge pairs is orthogonal and the edge lengths differ, a symmetric random walk in address coordinates becomes an anisotropic cloud in physical coordinates.

## 1.3 The 27-node Cell3

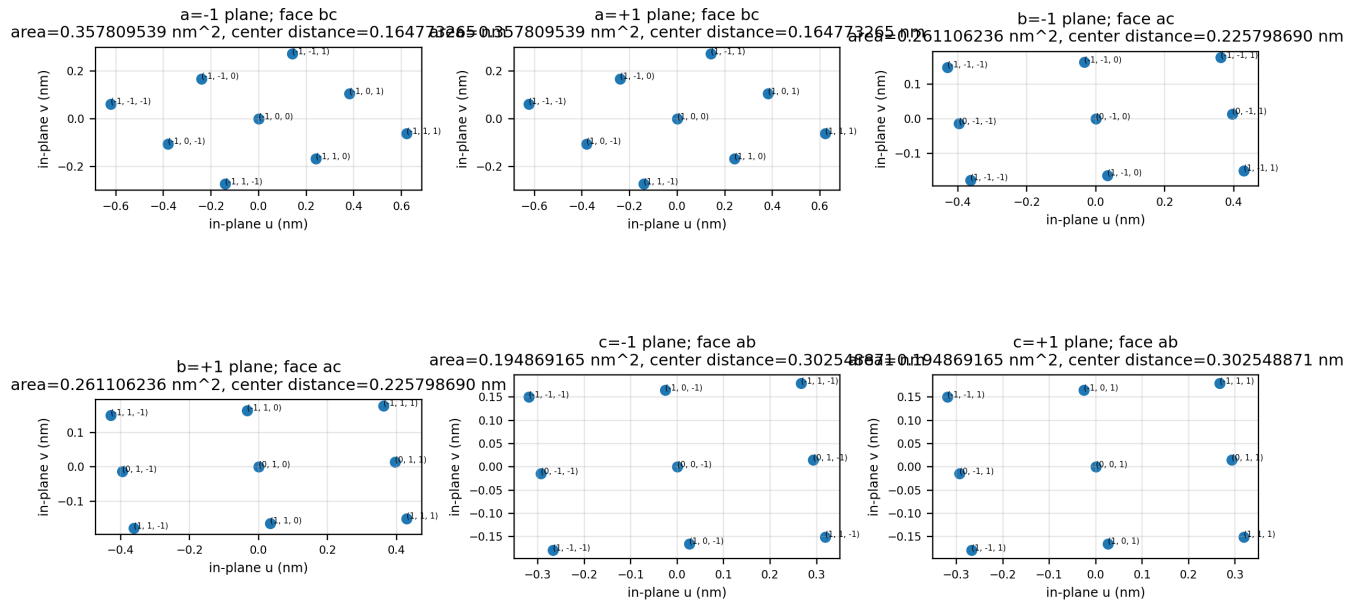
The locked physical Cell3 contains 27 distinguished half-step nodes. Each node has local integer label  $q$  in  $\{-1,0,1\}^3$  and physical position  $r(q)=Bq/2$ . The source ledger also supplies an address sum  $\lambda$ , polarity, and geometric node kind. The complete node registry is reproduced below so that the propagation geometry can be reconstructed without another document.

| Address      | $\lambda$ | polarity | kind        | x (nm)          | y (nm)          | z (nm)          |
|--------------|-----------|----------|-------------|-----------------|-----------------|-----------------|
| (-1, -1, -1) | -3        | -1       | corner      | -0.176351350136 | 0.118682429905  | 0.637311498325  |
| (-1, -1, 0)  | -2        | 1        | edge center | -0.181416008950 | -0.076211421278 | 0.292444294142  |
| (-1, -1, 1)  | -1        | -1       | corner      | -0.186480667763 | -0.271105272461 | -0.052422910042 |
| (-1, 0, -1)  | -2        | 1        | edge center | -0.158685962215 | 0.222342355125  | 0.363610955631  |
| (-1, 0, 0)   | -1        | -1       | face center | -0.163750621028 | 0.027448503942  | 0.018743751447  |

| Address     | lambda | polarity | kind        | x (nm)          | y (nm)          | z (nm)           |
|-------------|--------|----------|-------------|-----------------|-----------------|------------------|
| (-1, 0, 1)  | 0      | 1        | edge center | -0.168815279841 | -0.167445347241 | -0.326123452736  |
| (-1, 1, -1) | -1     | -1       | corner      | -0.141020574293 | 0.326002280345  | 0.089910412937   |
| (-1, 1, 0)  | 0      | 1        | edge center | -0.146085233106 | 0.131108429162  | -0.254956791247  |
| (-1, 1, 1)  | 1      | -1       | corner      | -0.151149891919 | -0.063785422021 | -0.599823995431  |
| (0, -1, -1) | -2     | 1        | edge center | -0.012600729109 | 0.091233925963  | 0.618567746878   |
| (0, -1, 0)  | -1     | -1       | face center | -0.017665387922 | -0.103659925220 | 0.273700542694   |
| (0, -1, 1)  | 0      | 1        | edge center | -0.022730046735 | -0.298553776403 | -0.071166661489  |
| (0, 0, -1)  | -1     | -1       | face center | 0.005064658813  | 0.194893851183  | 0.344867204184   |
| (0, 0, 0)   | 0      | 1        | body center | 0.000000000000  | 0.000000000000  | 0.000000000000   |
| (0, 0, 1)   | 1      | -1       | face center | -0.005064658813 | -0.194893851183 | -0.344867204184  |
| (0, 1, -1)  | 0      | 1        | edge center | 0.022730046735  | 0.298553776403  | 0.071166661489   |
| (0, 1, 0)   | 1      | -1       | face center | 0.017665387922  | 0.103659925220  | -0.273700542694  |
| (0, 1, 1)   | 2      | 1        | edge center | 0.012600729109  | -0.091233925963 | -0.618567746878  |
| (1, -1, -1) | -1     | -1       | corner      | 0.151149891919  | 0.063785422021  | 0.599823995431   |
| (1, -1, 0)  | 0      | 1        | edge center | 0.146085233106  | -0.131108429162 | 0.254956791247   |
| (1, -1, 1)  | 1      | -1       | corner      | 0.141020574293  | -0.326002280345 | -0.089910412937  |
| (1, 0, -1)  | 0      | 1        | edge center | 0.168815279841  | 0.167445347241  | 0.326123452736   |
| (1, 0, 0)   | 1      | -1       | face center | 0.163750621028  | -0.027448503942 | -0.018743751447  |
| (1, 0, 1)   | 2      | 1        | edge center | 0.158685962215  | -0.222342355125 | -0.363610955631  |
| (1, 1, -1)  | 1      | -1       | corner      | 0.186480667763  | 0.271105272461  | 0.052422910042   |
| (1, 1, 0)   | 2      | 1        | edge center | 0.181416008950  | 0.076211421278  | -0.292444294142  |
| (1, 1, 1)   | 3      | -1       | corner      | 0.176351350136  | -0.118682429905 | -0.6373111498325 |

## 1.4 Boundary planes and face geometry

Boundary-plane atlas: exact nine-node contents and physical in-plane dimensions



**Figure 4. Exact boundary-plane atlas.** Each of the six boundary planes is projected into its own physical in-plane coordinate system. Every plane contains its exact nine-node set, and the title records face family, area, and center-to-face distance.



| Channel | Coeff. | Face | Area (nm <sup>2</sup> ) | Opposite spacing (nm) | Center-face (nm)  | Nine node addresses                                                                                              |
|---------|--------|------|-------------------------|-----------------------|-------------------|------------------------------------------------------------------------------------------------------------------|
| a       | -1     | bc   | 0.357809539128562       | 0.329546529695447     | 0.164773264847723 | (-1, -1, -1); (-1, -1, 0); (-1, -1, 1); (-1, 0, -1); (-1, 0, 0); (-1, 0, 1); (-1, 1, -1); (-1, 1, 0); (-1, 1, 1) |
| a       | +1     | bc   | 0.357809539128562       | 0.329546529695447     | 0.164773264847723 | (1, -1, -1); (1, -1, 0); (1, -1, 1); (1, 0, -1); (1, 0, 0); (1, 0, 1); (1, 1, -1); (1, 1, 0); (1, 1, 1)          |
| b       | -1     | ac   | 0.261106235790205       | 0.451597379721285     | 0.225798689860642 | (-1, -1, -1); (-1, -1, 0); (-1, -1, 1); (0, -1, -1); (0, -1, 0); (0, -1, 1); (1, -1, -1); (1, -1, 0); (1, -1, 1) |
| b       | +1     | ac   | 0.261106235790205       | 0.451597379721285     | 0.225798689860642 | (-1, 1, -1); (-1, 1, 0); (-1, 1, 1); (0, 1, -1); (0, 1, 0); (0, 1, 1); (1, 1, -1); (1, 1, 0); (1, 1, 1)          |
| c       | -1     | ab   | 0.194869165460484       | 0.605097741518555     | 0.302548870759278 | (-1, -1, -1); (-1, 0, -1); (-1, 1, -1); (0, -1, -1); (0, 0, -1); (0, 1, -1); (1, -1, -1); (1, 0, -1); (1, 1, -1) |
| c       | +1     | ab   | 0.194869165460484       | 0.605097741518555     | 0.302548870759278 | (-1, -1, 1); (-1, 0, 1); (-1, 1, 1); (0, -1, 1); (0, 0, 1); (0, 1, 1); (1, -1, 1); (1, 0, 1); (1, 1, 1)          |

## 1.5 Corner-node lattice and selected initial state

A corner of one Cell3 is also a corner of neighboring cells in the volume-filling tiling. Translating a corner by any integer combination of the full edge vectors reaches an equivalent corner node. The experiment selects the local corner address  $q_0=(1,1,1)$ , whose physical position is

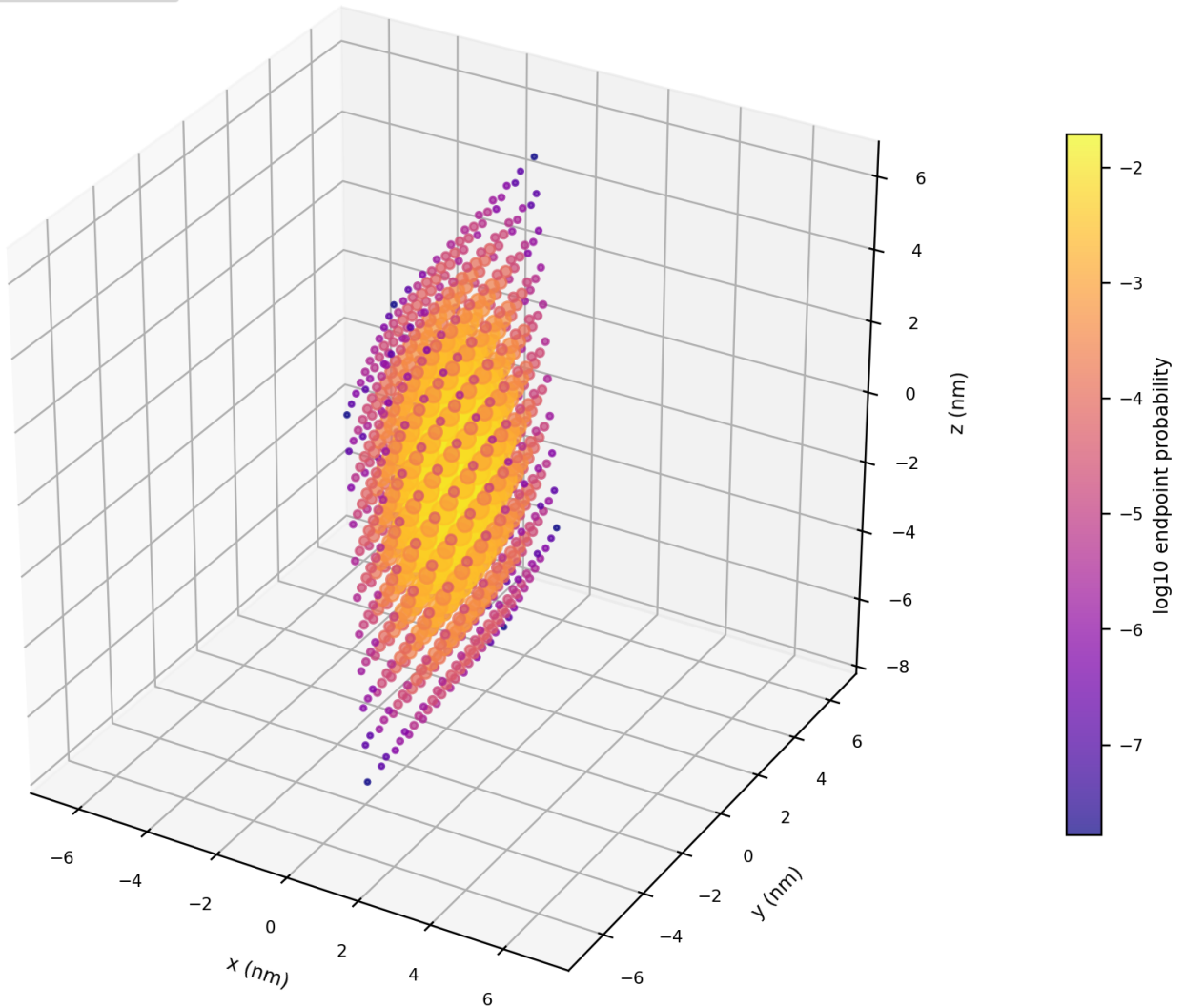
$$\begin{aligned} r_0 &= B(1,1,1)^T / 2 \\ &= (0.176351350136495, -0.118682429905037, -0.637311498325253) \text{ nm}. \end{aligned}$$

After a displacement address  $d$ , the endpoint corner is

$$r(d) = r_0 + B d.$$

## Volume-filling Cell3 corner lattice and exact 10-event destination support

★ selected (1,1,1) corner



**Figure 5. Volume-filling corner lattice and destination support.** A 3 by 3 by 3 neighborhood of exact Cell3 wireframes establishes the tiling around the selected corner. The full 891-node destination support is overlaid with marker size and color determined by exact endpoint probability.

## 2. Path history as a fiber over the destination address

### 2.1 Destination versus history

The endpoint address states where the system is after the event sequence. It does not state the order in which the transition channels were used. Every ordered sequence that ends at one destination belongs to the path-history fiber attached to that destination. In the unbiased ten-event experiment, the base is the finite set of 891 reachable destination addresses and the fiber over address  $d$  contains  $\Omega_{10}(d)$  ordered histories.

For a shortest path from one lattice address to another, the displacement components determine the required edge multiset. Reordering the same required  $+a$ ,  $-a$ ,  $+b$ ,  $-b$ ,  $+c$ , and  $-c$  moves produces multiple routes with the same combined edge length and the same endpoint. The source recursive example  $d=(4,-2,2)$  has eight required moves and  $8!/(4!2!2!)=420$  shortest histories. The ten-event propagation experiment is broader: it includes both shortest and non-shortest histories because every history is constrained by event count, not by minimum path length.

## 2.2 Four distinct measures of separation

| Quantity                | Meaning                                                                                                                      |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------|
| Event count $n$         | Number of transitions executed by the device. It is fixed at 10 in this experiment.                                          |
| Minimum edge-step count | $L1(d)= d_a + d_b + d_c $ . It is the fewest primitive edge transitions needed to reach the endpoint.                        |
| Traversed path length   | Sum of the physical lengths of every edge actually traversed by one history. It depends on how many a, b, and c edges occur. |
| Endpoint distance       | $R(d)=\sqrt{d^T G d}$ . It is the direct physical displacement from the starting corner to the endpoint.                     |

A ten-event history can return to the start, so  $n=10$  can coexist with minimum edge-step count zero and endpoint distance zero. Conversely, ten repeated +c moves yield the maximum endpoint distance and the maximum traversed length. The distinction is essential when interpreting entropy: the endpoint probability counts all histories of the declared event count, including histories that contain cancellation.

## 2.3 Fiber multiplicity, probability, and entropy

Let  $\Omega_{10}(d)$  denote the number of ten-event histories ending at  $d$ . Under the unbiased transition law, every ordered history has probability  $6^{-10}$ . Therefore

$$\begin{aligned} P_{10}(d) &= \Omega_{10}(d) / 6^{10}. \\ H_{\text{fiber}}(d) &= \ln \Omega_{10}(d). \\ I_{\text{endpoint}}(d) &= -\ln P_{10}(d) = 10 \ln 6 - \ln \Omega_{10}(d). \end{aligned}$$

The destination probability is thus a direct normalized measure of fiber size. A high-probability destination is high probability because many event histories resolve to it. A remote destination reached by only one sequence has a one-element fiber and zero fiber entropy.

## 3. Exact metastable-event model

### 3.1 Transition alphabet

One metastable resolution event is represented as a transition to one of six adjacent equivalent corner nodes. In integer address coordinates the transition alphabet is

$$S = \{(+1, 0, 0), (-1, 0, 0), (0, +1, 0), (0, -1, 0), (0, 0, +1), (0, 0, -1)\}.$$

The baseline model assigns probability  $1/6$  to each member of  $S$ . This expresses an unbiased device before any directional coupling, geometric asymmetry, drive asymmetry, or measurement selection has been introduced.

### 3.2 Exact recurrence

Let  $C_j(d)$  be the number of ordered  $j$ -event histories terminating at displacement  $d$ . The recurrence begins with one empty history at the origin and applies the six transitions at every event:

$$\begin{aligned} C_0(0, 0, 0) &= 1; \quad C_0(d) = 0 \text{ otherwise.} \\ C_{j+1}(d) &= \sum_{s \in S} C_j(d-s). \end{aligned}$$

After ten updates, the exact endpoint probability is  $C_{10}(d)/6^{10}$ . Integer arithmetic is used throughout the count recurrence, so there is no sampling error and no floating-point ambiguity in the history counts.

### 3.3 Generating function

The same endpoint counts are the coefficients of the Laurent polynomial

$$\begin{aligned} \Omega_{10}(d_a, d_b, d_c) &= [x^{d_a} y^{d_b} z^{d_c}] \\ & (x + x^{-1} + y + y^{-1} + z + z^{-1})^{10}. \end{aligned}$$

This equation makes the relationship among event histories, endpoint address, and probability explicit. Expanding the tenth power enumerates every ordered transition sequence. Collecting identical monomials groups those histories by their destination address.

### 3.4 Experiment assumptions

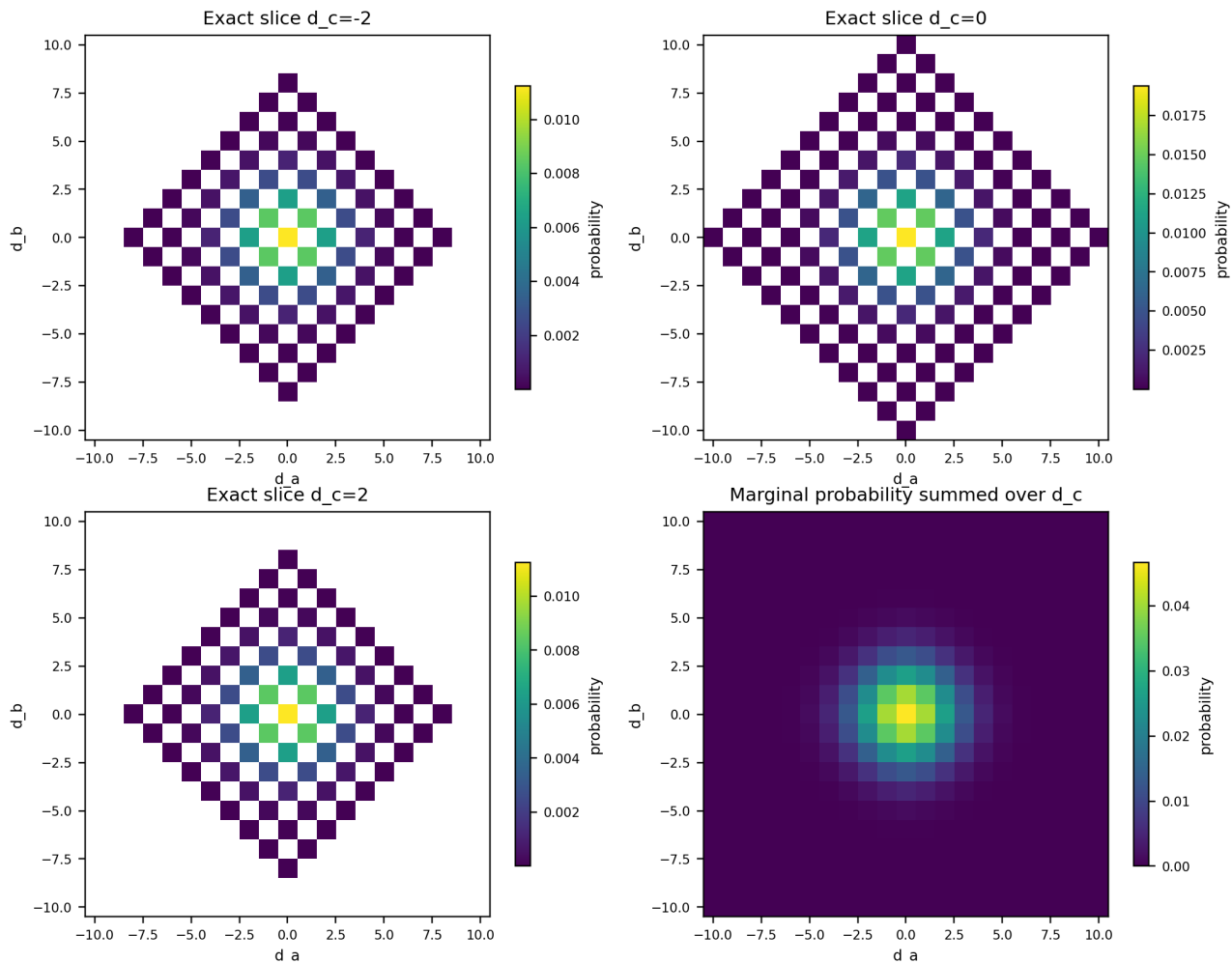
| Model element  | Declared value                                                                               |
|----------------|----------------------------------------------------------------------------------------------|
| Geometry       | Locked physical nanometre Cell3 basis and metric from the authoritative source file.         |
| Initial state  | Corner node q0=(1,1,1).                                                                      |
| Events         | Exactly 10 directed edge transitions.                                                        |
| Transition law | Independent and unbiased; each of six directed moves has probability 1/6.                    |
| Boundary       | Unbounded volume-filling Cell3 tiling; no reflection, absorption, or finite-device boundary. |
| Enumeration    | Exact dynamic programming over all 60,466,176 ordered histories.                             |
| Observation    | Final corner address and physical position after the tenth event.                            |

## 4. Exact destination distribution

### 4.1 Reachable support

The 60,466,176 histories collapse onto 891 destination nodes. Because the event count is even, every reachable displacement has even address parity and minimum edge-step count in {0,2,4,6,8,10}. The support contains one origin, 445 antipodal pairs, and therefore 446 unsigned physical radii under the source injectivity condition.

Exact integer-address probability slices after 10 events

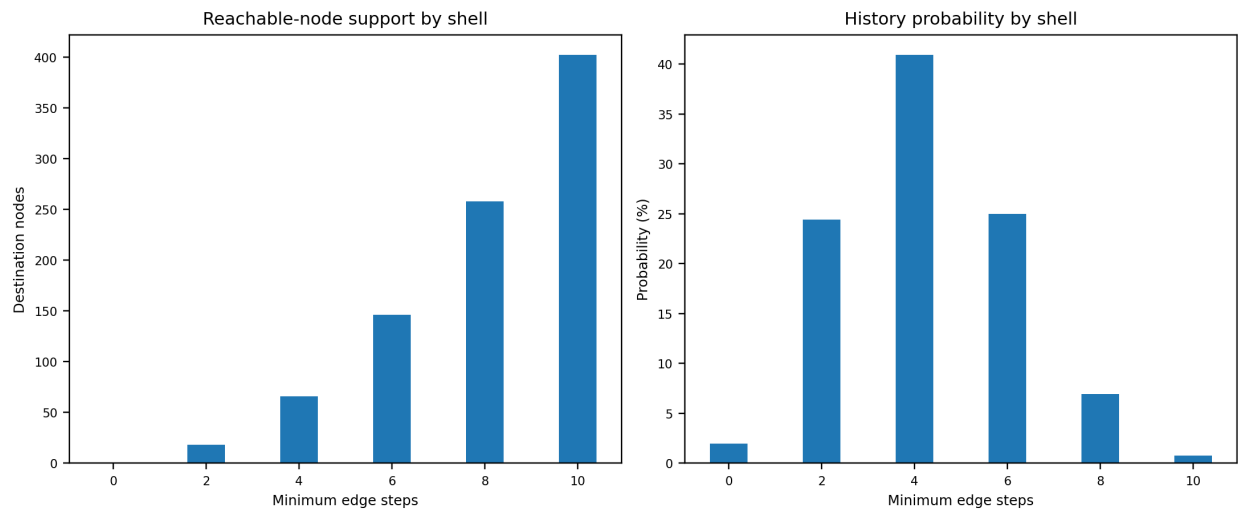


**Figure 6. Exact address-space probability slices.** Three fixed-d\_c sections and the d\_c-marginal display the discrete endpoint distribution in integer address coordinates. Empty checkerboard locations are forbidden by ten-event parity.

## 4.2 Minimum-step shells

The minimum edge-step count records the shortest return distance from the start to the endpoint in the primitive transition graph. It does not record how many events were actually executed; all histories here contain ten events. The exact shell distribution is

| Min. steps | Nodes | Histories  | Probability (%) | Mean R (nm) | Min R (nm)  | Max R (nm)  |
|------------|-------|------------|-----------------|-------------|-------------|-------------|
| 0          | 1     | 1,172,556  | 1.939193244     | 0.000000000 | 0.000000000 | 0.000000000 |
| 2          | 18    | 14,770,980 | 24.428500324    | 0.910344608 | 0.615518338 | 1.584639530 |
| 4          | 66    | 24,737,040 | 40.910541457    | 1.600727222 | 0.855955284 | 3.169279060 |
| 6          | 146   | 15,122,430 | 25.009734368    | 2.297185738 | 1.332794176 | 4.753918590 |
| 8          | 258   | 4,203,060  | 6.951092789     | 2.971265987 | 1.702907109 | 6.338558120 |
| 10         | 402   | 460,110    | 0.760937818     | 3.623015654 | 2.140966060 | 7.923197650 |



**Figure 7. Minimum-step shells.** The left panel counts reachable nodes; the right panel weights them by exact history probability. The four-step shell carries the greatest total probability even though the ten-step outer shell has the largest number of nodes.

The probability on the four-step shell is 40.910541 percent. The ten-step outer shell contains 402 nodes but only 0.760938 percent of the histories. Most histories partially cancel and therefore terminate inside the outer support boundary.

## 4.3 Return probability

Exactly 1,172,556 ordered histories return to the selected corner. The exact return probability is

$$\begin{aligned}
 P_{10}(0,0,0) &= 1,172,556 / 60,466,176 \\
 &= 0.019391932441701 \\
 &= 1.939193244 \text{ percent.}
 \end{aligned}$$

The starting corner is the single most probable destination. This concentration is entropic: the origin receives many balanced combinations of positive and negative channel transitions. No explicit central force is present in the null model.

## 4.4 Most probable endpoints

| Displacement d | Histories | Probability (%) | R (nm)         | Physical displacement (nm)                 |
|----------------|-----------|-----------------|----------------|--------------------------------------------|
| (0, 0, 0)      | 1,172,556 | 1.939193244     | 0.000000000000 | (0.000000000, 0.000000000, 0.000000000)    |
| (0, -1, 1)     | 890,820   | 1.473253410     | 0.615518338108 | (-0.045460093, -0.597107553, -0.142333323) |
| (0, 1, -1)     | 890,820   | 1.473253410     | 0.615518338108 | (0.045460093, 0.597107553, 0.142333323)    |
| (-1, 1, 0)     | 890,820   | 1.473253410     | 0.643531758134 | (-0.292170466, 0.262216858, -0.509913582)  |
| (1, -1, 0)     | 890,820   | 1.473253410     | 0.643531758134 | (0.292170466, -0.262216858, 0.509913582)   |
| (-1, -1, 0)    | 890,820   | 1.473253410     | 0.704964152883 | (-0.362832018, -0.152422843, 0.584888588)  |
| (1, 1, 0)      | 890,820   | 1.473253410     | 0.704964152883 | (0.362832018, 0.152422843, -0.584888588)   |
| (-1, 0, 1)     | 890,820   | 1.473253410     | 0.807200221618 | (-0.337630560, -0.334890694, -0.652246905) |
| (1, 0, -1)     | 890,820   | 1.473253410     | 0.807200221618 | (0.337630560, 0.334890694, 0.652246905)    |
| (-1, 0, -1)    | 890,820   | 1.473253410     | 0.909571953265 | (-0.317371924, 0.444684710, 0.727221911)   |
| (1, 0, 1)      | 890,820   | 1.473253410     | 0.909571953265 | (0.317371924, -0.444684710, -0.727221911)  |
| (0, -1, -1)    | 890,820   | 1.473253410     | 1.250773304957 | (-0.025201458, 0.182467852, 1.237135494)   |
| (0, 1, 1)      | 890,820   | 1.473253410     | 1.250773304957 | (0.025201458, -0.182467852, -1.237135494)  |
| (-2, 0, 0)     | 680,190   | 1.124909900     | 0.668359358118 | (-0.655002484, 0.109794016, 0.074975006)   |
| (2, 0, 0)      | 680,190   | 1.124909900     | 0.668359358118 | (0.655002484, -0.109794016, -0.074975006)  |
| (0, -2, 0)     | 680,190   | 1.124909900     | 1.172821780822 | (-0.070661552, -0.414639701, 1.094802171)  |
| (0, 2, 0)      | 680,190   | 1.124909900     | 1.172821780822 | (0.070661552, 0.414639701, -1.094802171)   |
| (0, 0, -2)     | 680,190   | 1.124909900     | 1.584639530085 | (0.020258635, 0.779575405, 1.379468817)    |
| (0, 0, 2)      | 680,190   | 1.124909900     | 1.584639530085 | (-0.020258635, -0.779575405, -1.379468817) |
| (-2, -1, 1)    | 509,040   | 0.841859092     | 0.855955284386 | (-0.700462578, -0.487313537, -0.067358317) |
| (2, 1, -1)     | 509,040   | 0.841859092     | 0.855955284386 | (0.700462578, 0.487313537, 0.067358317)    |
| (-2, 1, -1)    | 509,040   | 0.841859092     | 0.958370838138 | (-0.609542391, 0.706901569, 0.217308329)   |
| (2, -1, 1)     | 509,040   | 0.841859092     | 0.958370838138 | (0.609542391, -0.706901569, -0.217308329)  |
| (-1, -2, 1)    | 509,040   | 0.841859092     | 0.961433005793 | (-0.408292111, -0.749530395, 0.442555265)  |
| (1, 2, -1)     | 509,040   | 0.841859092     | 0.961433005793 | (0.408292111, 0.749530395, -0.442555265)   |

The twelve nearest two-step directional combinations each receive 890,820 histories, but their physical distances differ because the six face-diagonal combinations a+b, a-b, a+c, a-c, b+c, and b-c have different metric lengths. Equal combinatorial multiplicity therefore does not imply equal physical radius.

## 5. Physical distance and spatial distribution

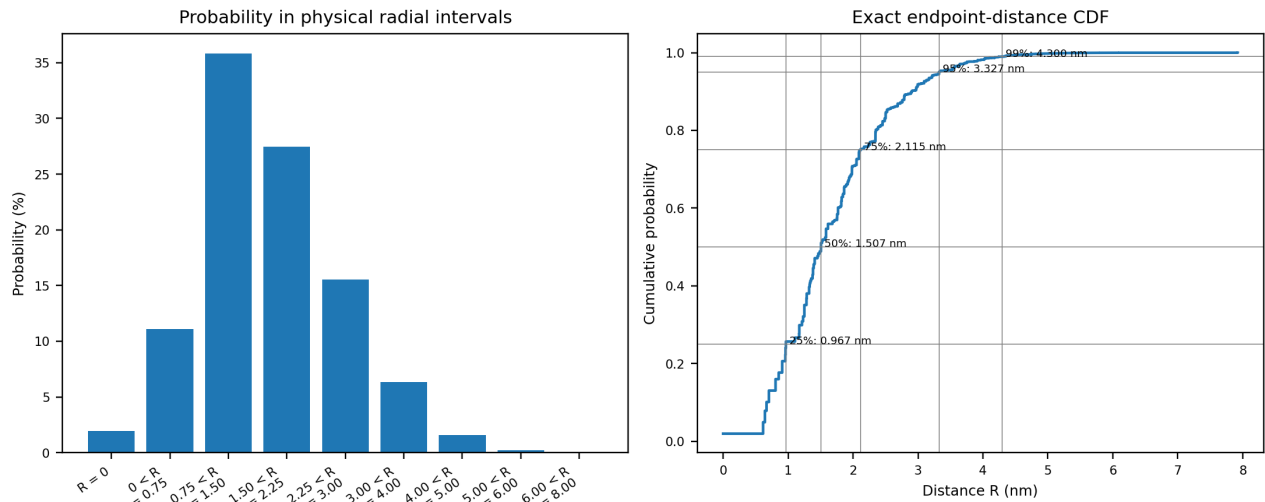
### 5.1 Exact radial statistics

| Statistic                          | Exact value                       |
|------------------------------------|-----------------------------------|
| Mean endpoint distance             | 1.685874112216074 nm              |
| Root-mean-square endpoint distance | 1.900284540418560 nm              |
| Mean squared endpoint distance     | 3.611081334553777 nm <sup>2</sup> |
| 25th percentile                    | 0.966653867094328 nm              |
| Median                             | 1.506561769554980 nm              |
| 75th percentile                    | 2.114892458649006 nm              |
| 95th percentile                    | 3.326832199052506 nm              |
| 99th percentile                    | 4.299673100420286 nm              |

The mean distance is an average of nonnegative radii and is therefore positive even though the mean vector displacement is zero. The root-mean-square distance is larger than the mean distance because it gives greater weight to the low-probability outer endpoints.

## 5.2 Radial probability intervals

| Physical interval       | Nodes | Histories  | Probability (%) |
|-------------------------|-------|------------|-----------------|
| $R = 0$ nm              | 1     | 1,172,556  | 1.939193244     |
| $0 < R \leq 0.75$ nm    | 8     | 6,705,300  | 11.089340262    |
| $0.75 < R \leq 1.50$ nm | 54    | 21,657,540 | 35.817611486    |
| $1.50 < R \leq 2.25$ nm | 134   | 16,602,990 | 27.458309915    |
| $2.25 < R \leq 3.00$ nm | 224   | 9,396,580  | 15.540225332    |
| $3.00 < R \leq 4.00$ nm | 206   | 3,845,114  | 6.359115549     |
| $4.00 < R \leq 5.00$ nm | 140   | 957,178    | 1.582997410     |
| $5.00 < R \leq 6.00$ nm | 84    | 122,222    | 0.202132842     |
| $6.00 < R \leq 8.00$ nm | 40    | 6,696      | 0.011073960     |



**Figure 8. Radial probability and exact cumulative distribution.** The left panel groups the complete discrete endpoint registry into declared physical intervals. The right panel is the exact probability-weighted CDF and marks the 25th, 50th, 75th, 95th, and 99th percentiles.

Approximately 76.3045 percent of the histories terminate within 2.25 nm of the starting corner. The outer tail beyond 4 nm contains only 1.7962 percent of the ensemble.

## 5.3 Extreme endpoints

The maximum physical displacement occurs when all ten events traverse the longest c edge with the same sign. The two antipodal endpoints  $d=(0,0,+10)$  and  $d=(0,0,-10)$  each have one history and therefore probability  $6^{-10}$ .

$$R_{\max} = 10 |c| = 7.923197650426852 \text{ nm.}$$

$$P_{\text{each}} = 1 / 60,466,176 = 1.653817168792020\text{e-}08.$$

The smallest nonzero endpoint displacement is  $d=(0, -1, 1)$  and its inverse. Its radius is 0.615518338107840 nm. Each member of the pair receives 890,820 histories and probability 1.473253410 percent. This result is a direct consequence of the short physical difference between two oblique basis vectors, not of a one-edge transition.

## 5.4 Mean and covariance

Positive and negative transitions are paired with equal probability, so the exact mean address displacement and mean physical displacement vanish within numerical roundoff:

$$E[d] = (0, 0, 0).$$

$$E[\Delta r] = (4.247\text{e-}18, 3.686\text{e-}18, -3.741\text{e-}18) \text{ nm.}$$

$$E[r_{10}] = r_0.$$

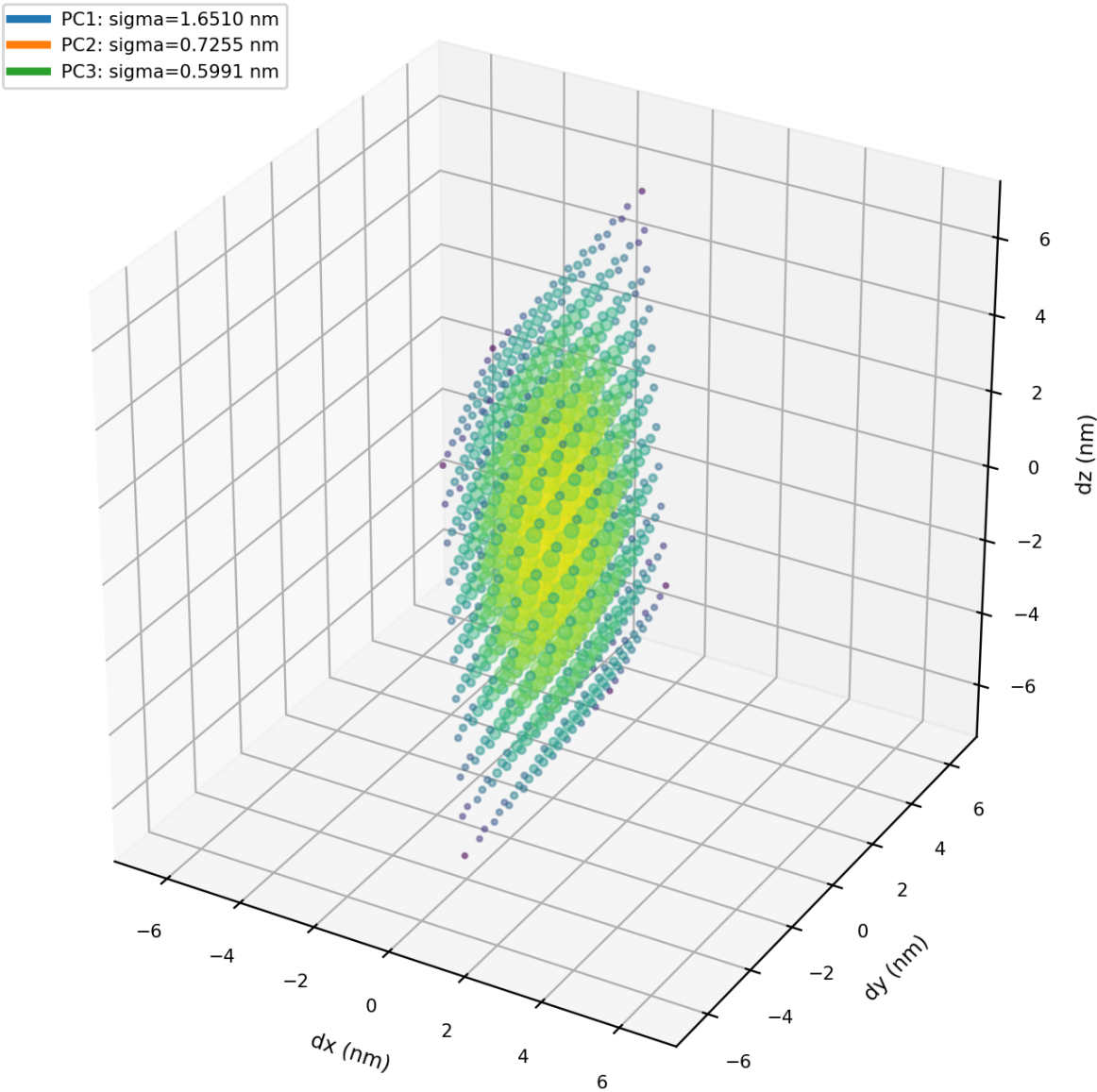
For one unbiased event, the physical increment covariance is  $(1/3)BB^T$  because each undirected edge family is chosen with total probability  $1/3$  and each sign has equal probability. Independent increments add, giving

$\text{Sigma}_{10} = (10/3) B B^T.$

| row | x                  | y                  | z                  |
|-----|--------------------|--------------------|--------------------|
| x   | 0.362026434484870  | -0.022352612196750 | -0.082102566345639 |
| y   | -0.022352612196750 | 0.659765515923760  | 0.524736102896161  |
| z   | -0.082102566345639 | 0.524736102896161  | 2.589289384145144  |

| Principal axis | Variance (nm^2)   | Std. dev. (nm)    | Unit direction                                     |
|----------------|-------------------|-------------------|----------------------------------------------------|
| 1              | 2.725821682437069 | 1.651006263597164 | (-0.035969282150, 0.246370427896, 0.968508039719)  |
| 2              | 0.526307530819477 | 0.725470558203072 | (-0.008496957347, 0.969024744456, -0.246817435257) |
| 3              | 0.358952121297228 | 0.599126131375713 | (0.999316772829, 0.017107217472, 0.032761725427)   |

Exact spatial covariance and one-standard-deviation principal ellipsoid



**Figure 9. Exact covariance ellipsoid.** The probability-weighted endpoint cloud is centered at the starting corner in displacement coordinates. The wireframe is the one-standard-deviation ellipsoid; the three heavy segments show its principal axes and exact standard deviations.



The longest principal spread is controlled by the long c edge and its strong nonorthogonal coupling with b. The distribution is inversion symmetric but not rotationally isotropic. Directional resolution is therefore already present in the geometry even before any directional probability bias is applied.

## 6. Traversed path length versus endpoint displacement

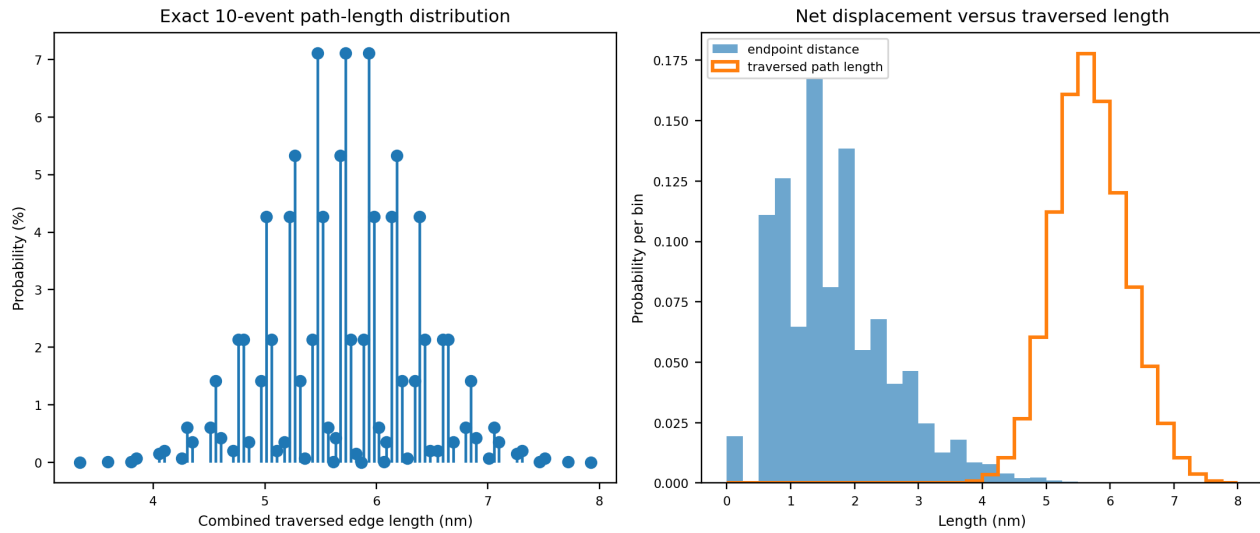
### 6.1 Ten events do not imply one physical path length

Every history contains ten events, but the physical edge families have unequal lengths. If a history contains  $n_a$  transitions along either sign of a,  $n_b$  along either sign of b, and  $n_c$  along either sign of c, then

$$n_a + n_b + n_c = 10,$$

$$L_{\text{path}} = n_a |a| + n_b |b| + n_c |c|.$$

The shortest possible ten-event traversed length is  $10|a|=3.341796790591105$  nm. The longest is  $10|c|=7.923197650426852$  nm. Across the unbiased ensemble, the mean traversed length is  $5.709701115043122$  nm and its standard deviation is  $0.592463207500103$  nm.



**Figure 10. Traversed path length and net endpoint distance.** The exact stem distribution follows the multinomial edge-family counts. The comparison panel shows that the path-length distribution is centered far above the endpoint-distance distribution because many transitions cancel in net displacement.

### 6.2 Expected traversed length

Each event selects an edge family with probability  $1/3$ . The expected edge length per event is therefore  $(|a|+|b|+|c|)/3$ , and the ten-event mean is

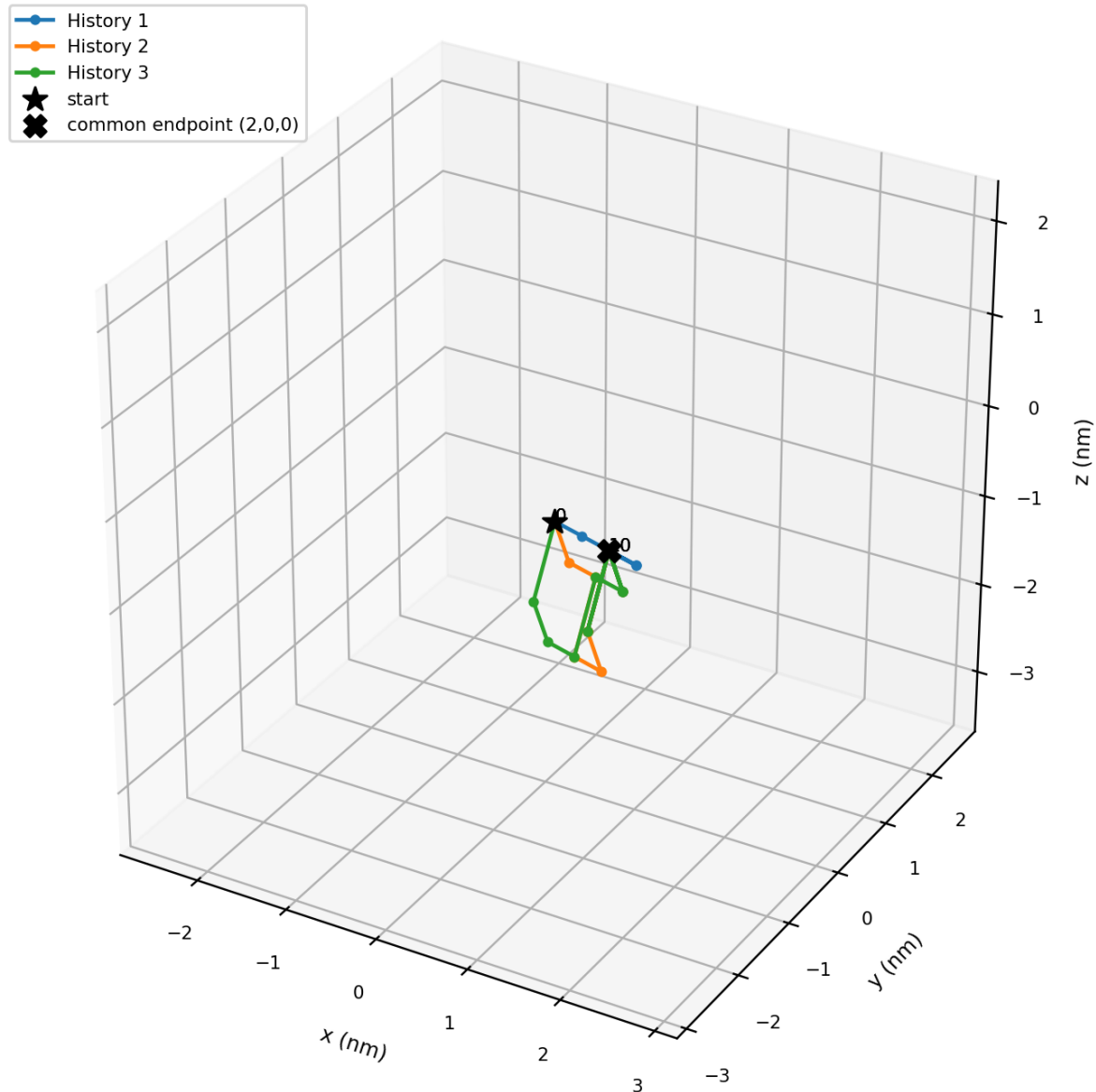
$$E[L_{\text{path}}] = 10 (|a|+|b|+|c|)/3$$

$$= 5.709701115043122 \text{ nm}.$$

The root-mean-square endpoint displacement is only  $1.900284540418560$  nm. The ratio  $E[L_{\text{path}}]/R_{\text{RMS}}$  is  $3.004656$ . This ratio quantifies the amount of traversed geometry that is removed by directional cancellation before the final endpoint is observed.

## 6.3 Same endpoint, different histories

Three distinct 10-event histories terminating at one destination



**Figure 11. Distinct histories over one endpoint.** Three declared ten-event sequences all terminate at displacement  $d=(2,0,0)$ . Their intermediate physical positions differ, so any phase, hysteresis, strain, polarization, or coherence accumulated along the route can differ even though the final address is identical.

**History 1:** +a, +a, +b, -b, +c, -c, +a, -a, +a, -a

**History 2:** +b, +a, +c, +a, -b, -c, +b, -b, +c, -c

**History 3:** +c, +b, +a, -c, +a, -b, +c, -c, +b, -b

This is the operational meaning of the path-history fiber in a metastable device. Endpoint occupancy measures the base address. History-sensitive observables probe information retained above that base.

## 7. Entropy of the exact propagation ensemble

### 7.1 History entropy

Every one of the 60,466,176 ordered histories has equal probability  $6^{-10}$ . The Shannon entropy of the complete event history is therefore

$$\begin{aligned} H_{\text{history}} &= -\sum_h 6^{-10} \ln(6^{-10}) \\ &= \ln(6^{10}) \\ &= 10 \ln 6 \\ &= 17.917594692280549 \text{ nats.} \end{aligned}$$

### 7.2 Endpoint entropy

When only the final destination is retained, histories terminating at the same address become indistinguishable. The endpoint entropy is

$$\begin{aligned} H_{\text{endpoint}} &= -\sum_d P_{10}(d) \ln P_{10}(d) \\ &= 5.364728188272016 \text{ nats.} \end{aligned}$$

The reachable support contains 891 nodes, but  $\exp(H_{\text{endpoint}})=213.733132$ . This entropy-equivalent support is much smaller because the endpoint distribution is strongly nonuniform.

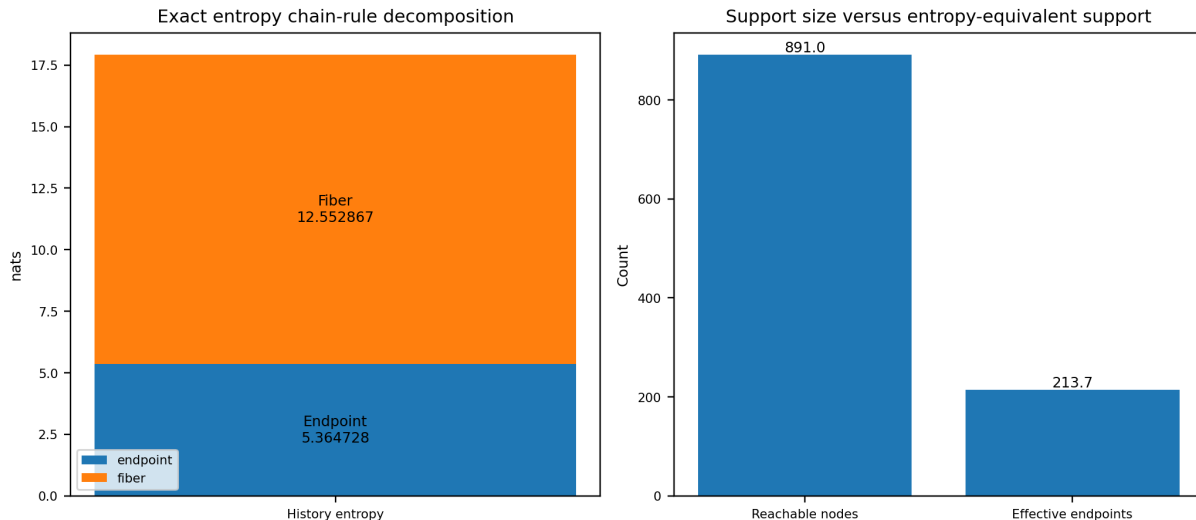
### 7.3 Conditional fiber entropy

For a known endpoint  $d$ , the conditional history distribution is uniform over its  $\Omega_{10}(d)$  compatible histories. Its entropy is  $\ln \Omega_{10}(d)$ . Averaging over endpoints gives

$$\begin{aligned} H_{\text{fiber}} &= \sum_d P_{10}(d) \ln \Omega_{10}(d) \\ &= 12.552866504008534 \text{ nats.} \end{aligned}$$

The exact chain rule is

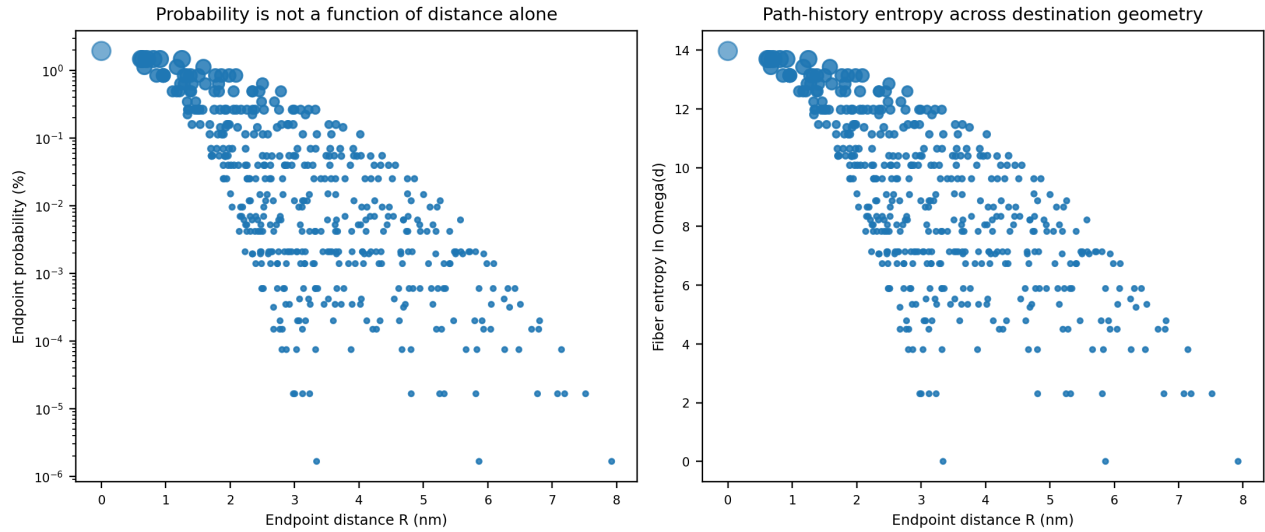
$$\begin{aligned} H_{\text{history}} &= H_{\text{endpoint}} + H_{\text{fiber}} \\ 17.917594692280549 &= 5.364728188272016 + 12.552866504008534. \end{aligned}$$



**Figure 12. Entropy decomposition.** The complete history entropy divides exactly into endpoint entropy and unresolved path-history fiber entropy. The right panel compares the 891-node support with the entropy-equivalent endpoint count.

The conditional fiber term contains 70.058882 percent of the total history entropy. Its exponential,  $\exp(H_{\text{fiber}})=282905.020591$ , is the entropy-equivalent number of histories remaining after an endpoint is observed.

## 7.4 Relationship between entropy and distance



**Figure 13. Distance, probability, and fiber entropy.** Each point is an exact destination. Multiple probabilities and fiber entropies occur at similar physical radii, demonstrating that Euclidean distance alone does not determine endpoint multiplicity. The complete displacement address does.

At fixed event count, the address  $d$  uniquely determines all of the following: endpoint position  $B d$ , endpoint distance  $\sqrt{d^T G d}$ , history count  $\Omega_{10}(d)$ , endpoint probability  $\Omega_{10}(d)/6^{10}$ , and fiber entropy  $\ln \Omega_{10}(d)$ . Entropy is therefore strictly address-resolved and geometrically embedded.

It is not, however, a monotonic scalar function of physical radius alone. Two different addresses can lie at comparable radii while admitting different numbers of histories. Likewise, antipodal addresses have equal radius and equal probability under the unbiased law. Directional resolution is preserved by the full address even when radius is used as a serialized projective key.

The exact entropic tendency toward the central region arises because balanced transition sequences are combinatorially numerous. This is an occupancy gradient generated by path multiplicity, not an imposed potential. In a metastable device, departures from this null gradient identify transition-law asymmetry or history-sensitive dynamics.

## 8. Metastable-device interpretation

### 8.1 Node occupancy as a metastable state distribution

In a metastable device, each corner node can represent a discrete resolved configuration and each directed Cell3 edge can represent an allowed transition channel between neighboring configurations. A sequence of resolution events produces a probability distribution over the accessible configuration nodes. The exact ten-event calculation provides the baseline occupancy distribution when all six channels are equally available and independent.

The concentration near the starting node is not evidence of a conventional restoring force by itself. It is the combinatorial consequence of the large number of histories that contain compensating transitions. The directional elongation of the cloud is not evidence of transition bias by itself. It is the physical image of an address-isotropic random walk under an anisotropic Cell3 metric.

### 8.2 Directional resolution as a deformed transition law

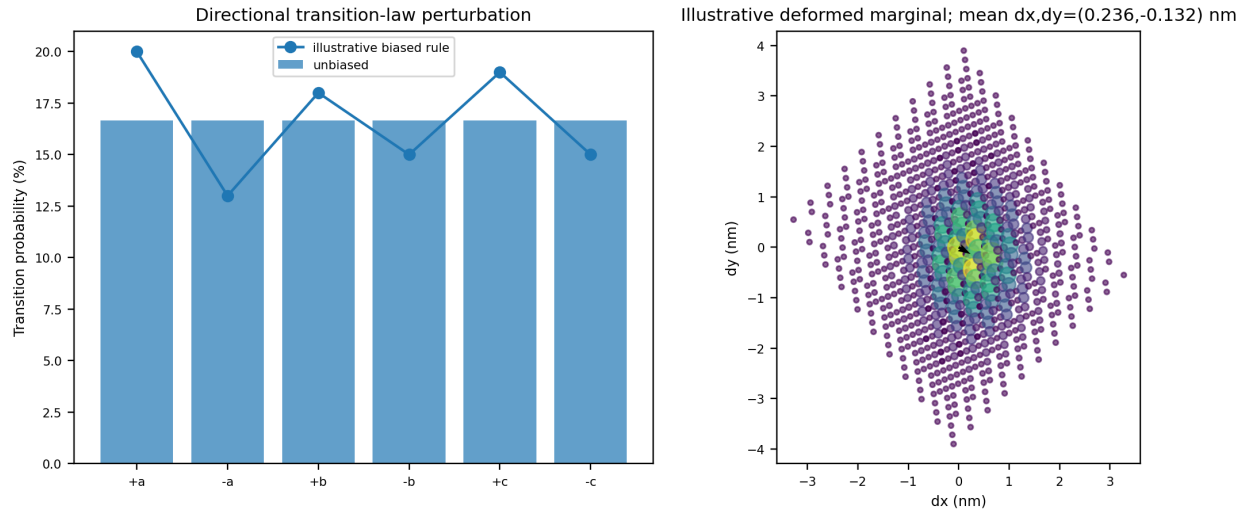
A directionally resolved device is represented by six probabilities

$$p_+(a+), p_-(a-), p_-(b+), p_-(b-), p_-(c+), p_-(c-), \\ \text{sum } p_s = 1.$$

The ten-event generating function becomes

$$(p_{-(a+)} x + p_{-(a-)} x^{-1} + p_{-(b+)} y + p_{-(b-)} y^{-1} + p_{-(c+)} z + p_{-(c-)} z^{-1})^{10}.$$

The coefficient of  $x^a y^b z^c$  is then the exact biased endpoint probability. This formula transforms the six directional event probabilities into a complete predicted spatial occupancy field.



**Figure 14. Directional-bias sensitivity calculation.** The declared probability vector (0.20, 0.13, 0.18, 0.15, 0.19, 0.15) is used only to demonstrate how a directional transition imbalance moves and deforms the endpoint distribution. It is a computed sensitivity case, not measured device data.

For the declared sensitivity vector, the exact ten-event mean displacement is (0.235798375, -0.132147031, -0.466355341) nm. In an experiment, the measured mean drift and covariance deformation can be inverted to estimate the signed channel imbalances.

### 8.3 First and second moments as device observables

For independent events with directional probabilities  $p_s$ , the mean address increment is

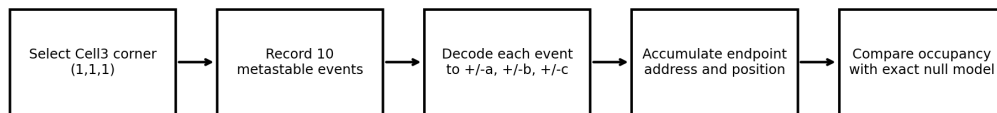
$$\mu_d = (p_{-(a+)} - p_{-(a-)}, p_{-(b+)} - p_{-(b-)}, p_{-(c+)} - p_{-(c-)}).$$

After  $n$  events, the mean physical displacement is

$$E[\Delta r_n] = n B \mu_d.$$

The covariance records both the total availability of each edge family and the signed drift. Together, mean and covariance provide a compact first diagnostic. The full endpoint histogram then tests higher-order structure, temporal dependence, non-Markov memory, and path-fiber effects that cannot be reconstructed from two moments alone.

### 8.4 Experimental protocol



Observable outputs: endpoint histogram, signed directional drift, covariance deformation, return rate, shell occupancy, and residual fiber-sensitive hysteresis

Null-model prediction: 60,466,176 equiprobable histories -> 891 destinations, with exact probabilities determined by path multiplicity.

**Figure 15. Metastable-device protocol.** The workflow begins with a prepared corner state, decodes ten transition events into the six Cell3 channels, accumulates exact endpoint addresses, and compares repeated trials against the exact null occupancy.

| Step                           | Required operation                                                                                              |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------|
| 1. State preparation           | Prepare or identify a repeatable metastable reference configuration assigned to one Cell3 corner node.          |
| 2. Event declaration           | Define one resolution event operationally before data collection, including timing and threshold rules.         |
| 3. Directional decoding        | Map each event to one of the six directed Cell3 edge channels using the device observables.                     |
| 4. Fixed-length trials         | Collect repeated trials containing exactly ten decoded events. Do not mix event counts.                         |
| 5. Endpoint reconstruction     | Sum the ten address increments and map the result to physical displacement $B_d$ .                              |
| 6. Null comparison             | Compare the observed 891-bin occupancy with the exact unbiased distribution, including parity-forbidden bins.   |
| 7. Directional inversion       | Estimate six transition probabilities or a lower-dimensional constrained model from the observed histogram.     |
| 8. Rotation and phase tests    | Repeat under controlled device orientation, phase reversal, and drive reversal to map signed changes.           |
| 9. Fiber-sensitive observables | At fixed endpoint, compare hysteresis, residual phase, relaxation, strain, or other history-sensitive readouts. |

## 8.5 Statistical comparisons

For  $M$  repeated ten-event trials, the destination counts follow a multinomial distribution under the declared model. A direct comparison can use the log-likelihood

$$\log L = \sum_d N_d \ln P_{10}(d),$$

where  $N_d$  is the observed number of trials at destination  $d$ . The model also predicts exact zeros at unreachable parity locations; occupancy at such a location indicates event miscount, channel-decoding error, boundary effects, or a transition outside the six-edge alphabet.

Useful compressed diagnostics include return probability, shell probabilities, mean drift, covariance eigenvalues and eigenvectors, radial CDF, and antipodal probability differences. These compressed tests should accompany, not replace, the full 891-bin likelihood because different directional laws can share the same first moments.

## 8.6 Detecting path-fiber memory

A memoryless endpoint model predicts only the destination distribution. A metastable device can retain additional path-dependent state. To detect that fiber information, group trials by identical endpoint and compare a secondary observable across distinct decoded histories. A statistically significant dependence on transition ordering at fixed endpoint is direct evidence that the device state is not exhausted by the endpoint address.

Candidate secondary observables include relaxation time, switching threshold, remanent polarization, residual mechanical strain, phase lag, emitted pulse shape, or subsequent transition bias. The specific observable must be selected from the device physics; the geometric role of the fiber is independent of that selection.

# 9. Worked interpretations

## 9.1 Why the four-step shell dominates

The outer ten-step shell contains histories with no net cancellation. Such histories are relatively rare because every event must contribute to the final minimum displacement. Interior shells admit many ways to insert canceling positive-negative pairs while preserving the endpoint. The four-step shell balances two effects: it has many available endpoint addresses and each endpoint has a large history fiber. Their product produces the maximum shell probability.

## 9.2 Why the origin is the most probable single endpoint

Returning to the origin requires each signed channel count to balance. Although this appears restrictive, the ten event positions can be assigned among the three edge families and their signs in many distinct orders. All such balanced words collect at the same address. No other single destination receives as many compatible histories.

### 9.3 Why the physical cloud is elongated

The transition law is symmetric in address coordinates, but the physical map is B. The c edge is approximately 2.37 times the length of a, and b and c have a substantial positive metric cross term. Consequently, equal address variance along the three primitive coordinates is transformed into unequal Cartesian variance. This is a geometric anisotropy, not a probability bias.

### 9.4 What a K-Field directional bias would look like

A signed imbalance  $p_{+}(a) \neq p_{-}(a)$  produces mean drift along physical vector a. Similar imbalances act along b and c. Changes in the sums  $p_{+}(a)+p_{-}(a)$ ,  $p_{+}(b)+p_{-}(b)$ , and  $p_{+}(c)+p_{-}(c)$  alter the covariance and therefore the ellipsoid dimensions. A directionally coupled device can thus show translation of the cloud, deformation of the cloud, or both.

### 9.5 Entropy and distance in the experimental record

For every endpoint, the machine-readable registry records distance, history count, probability, fiber entropy, and surprisal. Plotting fiber entropy against distance produces a structured cloud rather than a single curve. The experiment therefore supports the stronger address-resolved statement: geometry and entropy are deterministically linked through the discrete displacement address, while radial distance is one projection of that joint structure.

## 10. Findings and consequences

**Exact support.** Ten unbiased Cell3 edge events produce 60,466,176 equiprobable ordered histories and 891 distinct destination nodes.

**Entropic concentration.** The starting corner is the most probable single endpoint, with probability 1.939193244 percent, while the four-step shell contains 40.910541457 percent of the complete ensemble.

**Physical scale.** The mean distance is 1.685874112 nm and the RMS distance is 1.900284540 nm. The complete support extends from zero to 7.923197650 nm.

**Geometric anisotropy.** The exact endpoint covariance has principal standard deviations 1.651006264, 0.725470558, and 0.599126131 nm. The anisotropy arises from the locked Cell3 basis, even under an unbiased transition law.

**Information partition.** Endpoint observation retains 5.364728188 nats of the 17.917594692-nat history entropy. The remaining 12.552866504 nats reside in the unresolved path-history fiber.

**Metastable-device use.** The exact distribution is a null model for ten-event directional resolution. Mean drift, covariance deformation, shell redistribution, antipodal imbalance, and full-histogram likelihood provide direct tests of directional coupling. Secondary observables conditioned on endpoint provide a route to detect retained path-history information.

**Foundational conclusion.** Cell3 geometry defines the available transition network. Event history multiplicity defines the baseline probability of each destination. The physical basis converts address probability into an anisotropic spatial distribution. Directional device coupling appears as a controlled deformation of this exact geometric-entropy baseline.

## Appendix A. Complete exact formulas

| Object               | Formula                                               |
|----------------------|-------------------------------------------------------|
| Physical position    | $r(d)=r_0+B\ d$                                       |
| Metric distance      | $R(d)=\sqrt{d^T\ G\ d}$                               |
| Minimum edge steps   | $L1(d)= d_a + d_b + d_c $                             |
| Endpoint recurrence  | $C_{(j+1)}(d)=\sum_{(s\ in\ S)} C_j(d-s)$             |
| Endpoint probability | $P_{10}(d)=C_{10}(d)/6^{10}$                          |
| Fiber entropy        | $H_{fiber}(d)=\ln C_{10}(d)$                          |
| Endpoint surprisal   | $I(d)=10\ \ln 6-\ln C_{10}(d)$                        |
| Unbiased covariance  | $\Sigma_n=(n/3)BB^T$                                  |
| Biased mean          | $E[\Delta r_n]=nB(p_a+p_a^-, p_b+p_b^-, p_c+p_c^-)^T$ |
| Path length          | $L_{path}=n_a a +n_b b +n_c c $                       |
| Entropy chain rule   | $H_{history}=H_{endpoint}+E_d[\ln C_{10}(d)]$         |



## Appendix B. Full edge-family path-length ledger

| n_a | n_b | n_c | Path length (nm)  | Histories | Probability (%) |
|-----|-----|-----|-------------------|-----------|-----------------|
| 10  | 0   | 0   | 3.341796790591105 | 1,024     | 0.001693508781  |
| 9   | 1   | 0   | 3.594028001943135 | 10,240    | 0.016935087808  |
| 9   | 0   | 1   | 3.799936876574680 | 10,240    | 0.016935087808  |
| 8   | 2   | 0   | 3.846259213295165 | 46,080    | 0.076207895138  |
| 8   | 1   | 1   | 4.052168087926709 | 92,160    | 0.152415790276  |
| 7   | 3   | 0   | 4.098490424647196 | 122,880   | 0.203221053701  |
| 8   | 0   | 2   | 4.258076962558254 | 46,080    | 0.076207895138  |
| 7   | 2   | 1   | 4.304399299278741 | 368,640   | 0.609663161103  |
| 6   | 4   | 0   | 4.350721635999227 | 215,040   | 0.355636843977  |
| 7   | 1   | 2   | 4.510308173910285 | 368,640   | 0.609663161103  |
| 6   | 3   | 1   | 4.556630510630771 | 860,160   | 1.422547375908  |
| 5   | 5   | 0   | 4.602952847351256 | 258,048   | 0.426764212772  |
| 7   | 0   | 3   | 4.716217048541829 | 122,880   | 0.203221053701  |
| 6   | 2   | 2   | 4.762539385262315 | 1,290,240 | 2.133821063862  |
| 5   | 4   | 1   | 4.808861721982801 | 1,290,240 | 2.133821063862  |
| 4   | 6   | 0   | 4.855184058703287 | 215,040   | 0.355636843977  |
| 6   | 1   | 3   | 4.968448259893860 | 860,160   | 1.422547375908  |
| 5   | 3   | 2   | 5.014770596614346 | 2,580,480 | 4.267642127724  |
| 4   | 5   | 1   | 5.061092933334831 | 1,290,240 | 2.133821063862  |
| 3   | 7   | 0   | 5.107415270055316 | 122,880   | 0.203221053701  |
| 6   | 0   | 4   | 5.174357134525404 | 215,040   | 0.355636843977  |
| 5   | 2   | 3   | 5.220679471245890 | 2,580,480 | 4.267642127724  |
| 4   | 4   | 2   | 5.267001807966375 | 3,225,600 | 5.334552659656  |
| 3   | 6   | 1   | 5.313324144686862 | 860,160   | 1.422547375908  |
| 2   | 8   | 0   | 5.359646481407347 | 46,080    | 0.076207895138  |
| 5   | 1   | 4   | 5.426588345877434 | 1,290,240 | 2.133821063862  |
| 4   | 3   | 3   | 5.472910682597920 | 4,300,800 | 7.112736879541  |
| 3   | 5   | 2   | 5.519233019318406 | 2,580,480 | 4.267642127724  |
| 2   | 7   | 1   | 5.56555356038892  | 368,640   | 0.609663161103  |
| 1   | 9   | 0   | 5.611877692759379 | 10,240    | 0.016935087808  |
| 5   | 0   | 5   | 5.632497220508979 | 258,048   | 0.426764212772  |
| 4   | 2   | 4   | 5.678819557229464 | 3,225,600 | 5.334552659656  |
| 3   | 4   | 3   | 5.725141893949950 | 4,300,800 | 7.112736879541  |
| 2   | 6   | 2   | 5.771464230670436 | 1,290,240 | 2.133821063862  |
| 1   | 8   | 1   | 5.817786567390923 | 92,160    | 0.152415790276  |
| 0   | 10  | 0   | 5.864108904111408 | 1,024     | 0.001693508781  |
| 4   | 1   | 5   | 5.884728431861008 | 1,290,240 | 2.133821063862  |
| 3   | 3   | 4   | 5.931050768581494 | 4,300,800 | 7.112736879541  |
| 2   | 5   | 3   | 5.977373105301981 | 2,580,480 | 4.267642127724  |
| 1   | 7   | 2   | 6.023695442022467 | 368,640   | 0.609663161103  |
| 0   | 9   | 1   | 6.070017778742953 | 10,240    | 0.016935087808  |
| 4   | 0   | 6   | 6.090637306492553 | 215,040   | 0.355636843977  |
| 3   | 2   | 5   | 6.136959643213039 | 2,580,480 | 4.267642127724  |
| 2   | 4   | 4   | 6.183281979933525 | 3,225,600 | 5.334552659656  |
| 1   | 6   | 3   | 6.229604316654012 | 860,160   | 1.422547375908  |
| 0   | 8   | 2   | 6.275926653374497 | 46,080    | 0.076207895138  |

| n_a | n_b | n_c | Path length (nm)  | Histories | Probability (%) |
|-----|-----|-----|-------------------|-----------|-----------------|
| 3   | 1   | 6   | 6.342868517844583 | 860,160   | 1.422547375908  |
| 2   | 3   | 5   | 6.389190854565070 | 2,580,480 | 4.267642127724  |
| 1   | 5   | 4   | 6.435513191285555 | 1,290,240 | 2.133821063862  |
| 0   | 7   | 3   | 6.481835528006041 | 122,880   | 0.203221053701  |
| 3   | 0   | 7   | 6.548777392476127 | 122,880   | 0.203221053701  |
| 2   | 2   | 6   | 6.595099729196614 | 1,290,240 | 2.133821063862  |
| 1   | 4   | 5   | 6.641422065917100 | 1,290,240 | 2.133821063862  |
| 0   | 6   | 4   | 6.687744402637586 | 215,040   | 0.355636843977  |
| 2   | 1   | 7   | 6.801008603828159 | 368,640   | 0.609663161103  |
| 1   | 3   | 6   | 6.847330940548645 | 860,160   | 1.422547375908  |
| 0   | 5   | 5   | 6.893653277269130 | 258,048   | 0.426764212772  |
| 2   | 0   | 8   | 7.006917478459703 | 46,080    | 0.076207895138  |
| 1   | 2   | 7   | 7.053239815180189 | 368,640   | 0.609663161103  |
| 0   | 4   | 6   | 7.099562151900674 | 215,040   | 0.355636843977  |
| 1   | 1   | 8   | 7.259148689811733 | 92,160    | 0.152415790276  |
| 0   | 3   | 7   | 7.305471026532219 | 122,880   | 0.203221053701  |
| 1   | 0   | 9   | 7.465057564443278 | 10,240    | 0.016935087808  |
| 0   | 2   | 8   | 7.511379901163763 | 46,080    | 0.076207895138  |
| 0   | 1   | 9   | 7.717288775795308 | 10,240    | 0.016935087808  |
| 0   | 0   | 10  | 7.923197650426852 | 1,024     | 0.001693508781  |

# Appendix C. Top 100 endpoint registry

| d            | Histories | Probability           | R (nm)            | L1 | In Omega        | -In P          |
|--------------|-----------|-----------------------|-------------------|----|-----------------|----------------|
| (0, 0, 0)    | 1,172,556 | 1.939193244170096e-02 | 0.000000000000000 | 0  | 13.974696539356 | 3.942898152924 |
| (0, -1, 1)   | 890,820   | 1.473253410303307e-02 | 0.615518338107840 | 2  | 13.699897665842 | 4.217697026438 |
| (0, 1, -1)   | 890,820   | 1.473253410303307e-02 | 0.615518338107840 | 2  | 13.699897665842 | 4.217697026438 |
| (-1, 1, 0)   | 890,820   | 1.473253410303307e-02 | 0.643531758134457 | 2  | 13.699897665842 | 4.217697026438 |
| (1, -1, 0)   | 890,820   | 1.473253410303307e-02 | 0.643531758134457 | 2  | 13.699897665842 | 4.217697026438 |
| (-1, -1, 0)  | 890,820   | 1.473253410303307e-02 | 0.704964152883002 | 2  | 13.699897665842 | 4.217697026438 |
| (1, 1, 0)    | 890,820   | 1.473253410303307e-02 | 0.704964152883002 | 2  | 13.699897665842 | 4.217697026438 |
| (-1, 0, 1)   | 890,820   | 1.473253410303307e-02 | 0.807200221618470 | 2  | 13.699897665842 | 4.217697026438 |
| (1, 0, -1)   | 890,820   | 1.473253410303307e-02 | 0.807200221618470 | 2  | 13.699897665842 | 4.217697026438 |
| (-1, 0, -1)  | 890,820   | 1.473253410303307e-02 | 0.909571953264713 | 2  | 13.699897665842 | 4.217697026438 |
| (1, 0, 1)    | 890,820   | 1.473253410303307e-02 | 0.909571953264713 | 2  | 13.699897665842 | 4.217697026438 |
| (0, -1, -1)  | 890,820   | 1.473253410303307e-02 | 1.250773304957029 | 2  | 13.699897665842 | 4.217697026438 |
| (0, 1, 1)    | 890,820   | 1.473253410303307e-02 | 1.250773304957029 | 2  | 13.699897665842 | 4.217697026438 |
| (-2, 0, 0)   | 680,190   | 1.124909900040644e-02 | 0.668359358118221 | 2  | 13.430127449889 | 4.487467242392 |
| (2, 0, 0)    | 680,190   | 1.124909900040644e-02 | 0.668359358118221 | 2  | 13.430127449889 | 4.487467242392 |
| (0, -2, 0)   | 680,190   | 1.124909900040644e-02 | 1.172821780822282 | 2  | 13.430127449889 | 4.487467242392 |
| (0, 2, 0)    | 680,190   | 1.124909900040644e-02 | 1.172821780822282 | 2  | 13.430127449889 | 4.487467242392 |
| (0, 0, -2)   | 680,190   | 1.124909900040644e-02 | 1.584639530085370 | 2  | 13.430127449889 | 4.487467242392 |
| (0, 0, 2)    | 680,190   | 1.124909900040644e-02 | 1.584639530085370 | 2  | 13.430127449889 | 4.487467242392 |
| (-2, -1, 1)  | 509,040   | 8.418590916018900e-03 | 0.855955284386271 | 4  | 13.140281877907 | 4.777312814374 |
| (2, 1, -1)   | 509,040   | 8.418590916018900e-03 | 0.855955284386271 | 4  | 13.140281877907 | 4.777312814374 |
| (-2, 1, -1)  | 509,040   | 8.418590916018900e-03 | 0.958370838138188 | 4  | 13.140281877907 | 4.777312814374 |
| (2, -1, 1)   | 509,040   | 8.418590916018900e-03 | 0.958370838138188 | 4  | 13.140281877907 | 4.777312814374 |
| (-1, -2, 1)  | 509,040   | 8.418590916018900e-03 | 0.961433005793116 | 4  | 13.140281877907 | 4.777312814374 |
| (1, 2, -1)   | 509,040   | 8.418590916018900e-03 | 0.961433005793116 | 4  | 13.140281877907 | 4.777312814374 |
| (-1, 2, -1)  | 509,040   | 8.418590916018900e-03 | 0.966653867094328 | 4  | 13.140281877907 | 4.777312814374 |
| (1, -2, 1)   | 509,040   | 8.418590916018900e-03 | 0.966653867094328 | 4  | 13.140281877907 | 4.777312814374 |
| (-1, -1, 2)  | 509,040   | 8.418590916018900e-03 | 1.283252477468195 | 4  | 13.140281877907 | 4.777312814374 |
| (1, 1, -2)   | 509,040   | 8.418590916018900e-03 | 1.283252477468195 | 4  | 13.140281877907 | 4.777312814374 |
| (-2, 1, 1)   | 509,040   | 8.418590916018900e-03 | 1.323838290151039 | 4  | 13.140281877907 | 4.777312814374 |
| (2, -1, -1)  | 509,040   | 8.418590916018900e-03 | 1.323838290151039 | 4  | 13.140281877907 | 4.777312814374 |
| (-1, 1, -2)  | 509,040   | 8.418590916018900e-03 | 1.383977408983108 | 4  | 13.140281877907 | 4.777312814374 |
| (1, -1, 2)   | 509,040   | 8.418590916018900e-03 | 1.383977408983108 | 4  | 13.140281877907 | 4.777312814374 |
| (-2, -1, -1) | 509,040   | 8.418590916018900e-03 | 1.506561769554980 | 4  | 13.140281877907 | 4.777312814374 |
| (2, 1, 1)    | 509,040   | 8.418590916018900e-03 | 1.506561769554980 | 4  | 13.140281877907 | 4.777312814374 |
| (-1, 2, 1)   | 509,040   | 8.418590916018900e-03 | 1.769127703156480 | 4  | 13.140281877907 | 4.777312814374 |
| (1, -2, -1)  | 509,040   | 8.418590916018900e-03 | 1.769127703156480 | 4  | 13.140281877907 | 4.777312814374 |
| (-1, -2, -1) | 509,040   | 8.418590916018900e-03 | 1.863127595390463 | 4  | 13.140281877907 | 4.777312814374 |
| (1, 2, 1)    | 509,040   | 8.418590916018900e-03 | 1.863127595390463 | 4  | 13.140281877907 | 4.777312814374 |
| (-1, 1, 2)   | 509,040   | 8.418590916018900e-03 | 1.983692934780487 | 4  | 13.140281877907 | 4.777312814374 |
| (1, -1, -2)  | 509,040   | 8.418590916018900e-03 | 1.983692934780487 | 4  | 13.140281877907 | 4.777312814374 |
| (-1, -1, -2) | 509,040   | 8.418590916018900e-03 | 2.090305449782448 | 4  | 13.140281877907 | 4.777312814374 |
| (1, 1, 2)    | 509,040   | 8.418590916018900e-03 | 2.090305449782448 | 4  | 13.140281877907 | 4.777312814374 |
| (0, -2, 2)   | 383,040   | 6.334781283340954e-03 | 1.231036676215680 | 4  | 12.855894701352 | 5.061699990929 |
| (0, 2, -2)   | 383,040   | 6.334781283340954e-03 | 1.231036676215680 | 4  | 12.855894701352 | 5.061699990929 |
| (-2, 2, 0)   | 383,040   | 6.334781283340954e-03 | 1.287063516268914 | 4  | 12.855894701352 | 5.061699990929 |

| d            | Histories | Probability           | R (nm)            | L1 | In Omega        | -ln P          |
|--------------|-----------|-----------------------|-------------------|----|-----------------|----------------|
| (2, -2, 0)   | 383,040   | 6.334781283340954e-03 | 1.287063516268914 | 4  | 12.855894701352 | 5.061699990929 |
| (-2, -2, 0)  | 383,040   | 6.334781283340954e-03 | 1.409928305766004 | 4  | 12.855894701352 | 5.061699990929 |
| (2, 2, 0)    | 383,040   | 6.334781283340954e-03 | 1.409928305766004 | 4  | 12.855894701352 | 5.061699990929 |
| (-2, 0, 2)   | 383,040   | 6.334781283340954e-03 | 1.614400443236939 | 4  | 12.855894701352 | 5.061699990929 |
| (2, 0, -2)   | 383,040   | 6.334781283340954e-03 | 1.614400443236939 | 4  | 12.855894701352 | 5.061699990929 |
| (-2, 0, -2)  | 383,040   | 6.334781283340954e-03 | 1.819143906529426 | 4  | 12.855894701352 | 5.061699990929 |
| (2, 0, 2)    | 383,040   | 6.334781283340954e-03 | 1.819143906529426 | 4  | 12.855894701352 | 5.061699990929 |
| (0, -2, -2)  | 383,040   | 6.334781283340954e-03 | 2.501546609914058 | 4  | 12.855894701352 | 5.061699990929 |
| (0, 2, 2)    | 383,040   | 6.334781283340954e-03 | 2.501546609914058 | 4  | 12.855894701352 | 5.061699990929 |
| (-3, 1, 0)   | 296,520   | 4.903898668902098e-03 | 1.106661761232222 | 4  | 12.599869948771 | 5.317724743510 |
| (3, -1, 0)   | 296,520   | 4.903898668902098e-03 | 1.106661761232222 | 4  | 12.599869948771 | 5.317724743510 |
| (-3, 0, 1)   | 296,520   | 4.903898668902098e-03 | 1.170141752337907 | 4  | 12.599869948771 | 5.317724743510 |
| (3, 0, -1)   | 296,520   | 4.903898668902098e-03 | 1.170141752337907 | 4  | 12.599869948771 | 5.317724743510 |
| (-3, -1, 0)  | 296,520   | 4.903898668902098e-03 | 1.213764496572903 | 4  | 12.599869948771 | 5.317724743510 |
| (3, 1, 0)    | 296,520   | 4.903898668902098e-03 | 1.213764496572903 | 4  | 12.599869948771 | 5.317724743510 |
| (-3, 0, -1)  | 296,520   | 4.903898668902098e-03 | 1.377126915617825 | 4  | 12.599869948771 | 5.317724743510 |
| (3, 0, 1)    | 296,520   | 4.903898668902098e-03 | 1.377126915617825 | 4  | 12.599869948771 | 5.317724743510 |
| (0, -3, 1)   | 296,520   | 4.903898668902098e-03 | 1.394386477216142 | 4  | 12.599869948771 | 5.317724743510 |
| (0, 3, -1)   | 296,520   | 4.903898668902098e-03 | 1.394386477216142 | 4  | 12.599869948771 | 5.317724743510 |
| (-1, 3, 0)   | 296,520   | 4.903898668902098e-03 | 1.755651915884096 | 4  | 12.599869948771 | 5.317724743510 |
| (1, -3, 0)   | 296,520   | 4.903898668902098e-03 | 1.755651915884096 | 4  | 12.599869948771 | 5.317724743510 |
| (-1, -3, 0)  | 296,520   | 4.903898668902098e-03 | 1.825058259101547 | 4  | 12.599869948771 | 5.317724743510 |
| (1, 3, 0)    | 296,520   | 4.903898668902098e-03 | 1.825058259101547 | 4  | 12.599869948771 | 5.317724743510 |
| (0, -1, 3)   | 296,520   | 4.903898668902098e-03 | 2.053157731232385 | 4  | 12.599869948771 | 5.317724743510 |
| (0, 1, -3)   | 296,520   | 4.903898668902098e-03 | 2.053157731232385 | 4  | 12.599869948771 | 5.317724743510 |
| (-1, 0, 3)   | 296,520   | 4.903898668902098e-03 | 2.344778910263054 | 4  | 12.599869948771 | 5.317724743510 |
| (1, 0, -3)   | 296,520   | 4.903898668902098e-03 | 2.344778910263054 | 4  | 12.599869948771 | 5.317724743510 |
| (0, -3, -1)  | 296,520   | 4.903898668902098e-03 | 2.345426774679081 | 4  | 12.599869948771 | 5.317724743510 |
| (0, 3, 1)    | 296,520   | 4.903898668902098e-03 | 2.345426774679081 | 4  | 12.599869948771 | 5.317724743510 |
| (-1, 0, -3)  | 296,520   | 4.903898668902098e-03 | 2.454635402492401 | 4  | 12.599869948771 | 5.317724743510 |
| (1, 0, 3)    | 296,520   | 4.903898668902098e-03 | 2.454635402492401 | 4  | 12.599869948771 | 5.317724743510 |
| (0, -1, -3)  | 296,520   | 4.903898668902098e-03 | 2.787861147341737 | 4  | 12.599869948771 | 5.317724743510 |
| (0, 1, 3)    | 296,520   | 4.903898668902098e-03 | 2.787861147341737 | 4  | 12.599869948771 | 5.317724743510 |
| (-2, -2, 2)  | 207,900   | 3.438285893918610e-03 | 1.332794175875419 | 6  | 12.244812473846 | 5.672782218434 |
| (2, 2, -2)   | 207,900   | 3.438285893918610e-03 | 1.332794175875419 | 6  | 12.244812473846 | 5.672782218434 |
| (-2, 2, -2)  | 207,900   | 3.438285893918610e-03 | 1.465595696055792 | 6  | 12.244812473846 | 5.672782218434 |
| (2, -2, 2)   | 207,900   | 3.438285893918610e-03 | 1.465595696055792 | 6  | 12.244812473846 | 5.672782218434 |
| (-2, 2, 2)   | 207,900   | 3.438285893918610e-03 | 2.487420174828151 | 6  | 12.244812473846 | 5.672782218434 |
| (2, -2, -2)  | 207,900   | 3.438285893918610e-03 | 2.487420174828151 | 6  | 12.244812473846 | 5.672782218434 |
| (-2, -2, -2) | 207,900   | 3.438285893918610e-03 | 2.687307243351850 | 6  | 12.244812473846 | 5.672782218434 |
| (2, 2, 2)    | 207,900   | 3.438285893918610e-03 | 2.687307243351850 | 6  | 12.244812473846 | 5.672782218434 |
| (-3, -2, 1)  | 158,760   | 2.625600137174211e-03 | 1.344505713508408 | 6  | 11.975148906897 | 5.942445785384 |
| (3, 2, -1)   | 158,760   | 2.625600137174211e-03 | 1.344505713508408 | 6  | 11.975148906897 | 5.942445785384 |
| (-3, 2, -1)  | 158,760   | 2.625600137174211e-03 | 1.355689653304492 | 6  | 11.975148906897 | 5.942445785384 |
| (3, -2, 1)   | 158,760   | 2.625600137174211e-03 | 1.355689653304492 | 6  | 11.975148906897 | 5.942445785384 |
| (-1, -3, 2)  | 158,760   | 2.625600137174211e-03 | 1.452397342844168 | 6  | 11.975148906897 | 5.942445785384 |
| (1, 3, -2)   | 158,760   | 2.625600137174211e-03 | 1.452397342844168 | 6  | 11.975148906897 | 5.942445785384 |
| (-1, 3, -2)  | 158,760   | 2.625600137174211e-03 | 1.487424594022597 | 6  | 11.975148906897 | 5.942445785384 |

| d           | Histories | Probability           | R (nm)            | L1 | ln Omega        | -ln P          |
|-------------|-----------|-----------------------|-------------------|----|-----------------|----------------|
| (1, -3, 2)  | 158,760   | 2.625600137174211e-03 | 1.487424594022597 | 6  | 11.975148906897 | 5.942445785384 |
| (-3, -1, 2) | 158,760   | 2.625600137174211e-03 | 1.507145924072824 | 6  | 11.975148906897 | 5.942445785384 |
| (3, 1, -2)  | 158,760   | 2.625600137174211e-03 | 1.507145924072824 | 6  | 11.975148906897 | 5.942445785384 |
| (-2, 3, -1) | 158,760   | 2.625600137174211e-03 | 1.522577689461215 | 6  | 11.975148906897 | 5.942445785384 |
| (2, -3, 1)  | 158,760   | 2.625600137174211e-03 | 1.522577689461215 | 6  | 11.975148906897 | 5.942445785384 |
| (-2, -3, 1) | 158,760   | 2.625600137174211e-03 | 1.569647393018519 | 6  | 11.975148906897 | 5.942445785384 |

The machine-readable companion contains the complete 891-endpoint registry with physical coordinates, distances, exact integer history counts, probabilities, fiber entropies, and endpoint surprisals.

## Appendix D. Reproducible exact algorithm

The following language-independent pseudocode reproduces the endpoint counts. It uses an integer map from displacement triples to history counts.

```
counts = {(0,0,0): 1}
repeat 10 times:
    next = empty integer map
    for each (d,count) in counts:
        for each s in {+a, -a,+b, -b,+c, -c}:
            next[d+s] += count
    counts = next
for each endpoint d:
    probability[d] = counts[d] / 6^10
    position[d] = r0 + B d
    distance[d] = sqrt(d^T G d)
```

## Appendix E. Archival record

| Field                      | Record                                                                                                          |
|----------------------------|-----------------------------------------------------------------------------------------------------------------|
| Document title             | K-Field 10-Event Cell3 Propagation Experiment                                                                   |
| Principal Investigator     | P. Dwayne Esterline                                                                                             |
| Credentials                | BS, Huntington University (1999); MBA, Indiana Wesleyan University (2008)                                       |
| Organization               | Krytronx                                                                                                        |
| Classification             | Proprietary Confidential Information and Trade Secret of Krytronx, LLC. All Rights Reserved. Do Not Distribute. |
| Generation UTC             | 2026-07-28T02:12:45Z                                                                                            |
| Generation epoch           | 1785204765                                                                                                      |
| Output PDF                 | K-Field_10_event_Cell3_propagation_experiment_1785204765.pdf                                                    |
| Machine-readable companion | K-Field_10_event_Cell3_propagation_experiment_1785204765.json                                                   |
| Report generation          | Python ReportLab; built-in core fonts only                                                                      |
| Figure generation          | Python numerical computation and raster rendering from the exact experiment ledger                              |
| Source file                | K-Field_overdetermined_one-dimensional_serialized_spectral_address_space_1785179337(1).json                     |
| Source SHA-256             | 34ccd7ee5c1e7a9071d4cf1ef7164620e067eba133fe33ce43044d0c1556e51b                                                |
| Coordinate frame           | Physical Cell3 basis as supplied; Cartesian components labeled ICRS in source                                   |
| Units                      | Nanometres for physical geometry; natural logarithms for entropy                                                |

**P. Dwayne Esterline**  
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